Honest Inference for Tensor Contrasts under Tucker and Tensor-Train Rank Bounds

Theory-reviewed draft: September 9, 2026

Abstract

We study honest confidence intervals for prescribed orthogonal contrasts of a Gaussian tensor mean with bounded Frobenius norm and bounded Tucker or tensor-train ranks. The actual ranks are unknown and nonzero singular values may be arbitrarily small. Canonical tensor expansions imply an $\ell_{2/K}$ bound on the relevant diagonal, which calibrates a soft-threshold interval without rank recovery. For equal mode dimension p, fixed order $K \geq 3$, fixed rank caps and norm bound $M = \tau\sqrt{p}$, the interval has length $O\{\tau(\log p/p)^{(K-2)/(2K)}\}$, where τ is the coordinate noise standard deviation. We prove that this is smaller in order than every fixed-radius affine interval on the same class and than the expected length at zero of every interval honest on a single-unfolding relaxation. At rank one, the diagonal image is exactly an $\ell_{2/K}$ ball. A confidence-interval specialization of classical sparse-sequence theory shows that the displayed rate is optimal when only that diagonal is used. A finite, computationally expensive scan establishes the full-data optimal expected length at zero, of order $\tau\sqrt{\log p/p}$, for fixed $K \geq 4$. In order three, a harmonic sign mixture disproves this rate and leaves a factor $\sqrt{\log p}$ between new lower and upper bounds. Full-data worst-case optimality remains open. In 172 Gaussian simulation configurations with 2,000 replications each, empirical coverage of the selected interval ranges from 99.75% to 100%, demonstrating conservative calibration. The results quantify the inferential value of joint rank constraints while distinguishing rank-cap honesty, diagonal-information optimality and unresolved adaptation to actual rank.

Keywords: honest confidence intervals; linear functionals; Gaussian sequence model; tensor train; Tucker rank; nonconvex parameter space.

1 Introduction

Tensor rank models encode restrictions on several unfoldings of the same parameter. For statistical inference, these restrictions are useful only if their effect on interval length survives uncertainty about the actual ranks. A component can be too weak to identify reliably while still belonging to the class over which coverage is required. Consider a third-order $p \times p \times p$ mean tensor A with $\|A\|_F \leq 1$ and rank at most one. From $n = p$ independent full tensors with standard Gaussian errors, the sample mean has noise level $\tau = p^{-1/2}$. The normalized diagonal sum $p^{-1/2}\sum_i A_{iii}$ has a direct interval of length proportional to τ. Can the simultaneous rank restrictions provide a smaller interval with coverage at every admissible mean, including zero?

Existing theory treats several aspects of this question. Tensor functional inference on regular low-rank strata uses accurate subspace estimation and suitable signal conditions [Xu et al., 2025]. Matrix inference without exact-rank estimation is available under quantitative strength and alignment conditions [Choi et al., 2026]. Classical work establishes nonlinear functional estimation and the limits of honest adaptation on nonconvex parameter sets [Cai and Low, 2004b,a, Collier et al., 2017].

These precedents motivate a narrower comparison: determine whether joint tensor rank constraints permit a length improvement that cannot be retained by intervals required to cover a larger model defined by a single unfolding.

We study this comparison in the fully observed Gaussian mean experiment, with a known noise variance, a supplied Frobenius bound and supplied upper rank bounds. The loading is a sum of orthogonal product tensors. Honesty means a finite-sample lower bound on coverage over the entire rank-cap class; it includes every lower-rank stratum and every arbitrarily small nonzero singular value. For fixed $K \geq 3$, equal mode size p and $M = \tau\sqrt{p}$, our interval has length $o(\tau)$. By comparison, any honest fixed-radius affine interval on the same class has length at least a constant times τ, and any interval honest over a single-unfolding relaxation has expected length at zero at least a constant times τ. Matricization preserves the data. The latter comparison concerns the parameter space over which coverage is required.

Three mathematical issues govern the argument. Uniform control at rank boundaries requires an envelope that avoids inverses of small singular values. A lower bound for the enlarged model must apply to arbitrary interval procedures, including those using all tensor observations. Finally, optimality depends on which observations a procedure uses: the coordinate diagonal and the full tensor contain different statistical information. This distinction prevents a sharp result for a Gaussian sequence marginal from being mistaken for a sharp full-tensor result.

Canonical Tucker and TT expansions control the sum of squared component weights and hence the $\ell_{2/K}$ size of an orthogonal diagonal. This exponent is familiar from tensor factor geometry [Fan et al., 2023]. A deterministic error envelope for classical soft thresholding [Donoho and Johnstone, 1994] then produces an honest interval. Common-target Gaussian mixtures provide the comparison lower bounds. In the third-order example, the radius is at most $\tau\{8\log(2p/\alpha)/p\}^{1/6}$. At rank one, an exact diagonal-image identity permits a specialization of sparse-sequence theory [Collier et al., 2017]; its lower bound matches this order for diagonal procedures, while full-data analysis requires a different argument. A scan of coordinate subtensors exploits off-diagonal information. A sign-coupled harmonic mixture shows that the cubic case has a larger zero-point lower rate than orders four and above.

The paper develops the following results:

1. An explicit diagonal envelope for Tucker and TT cap classes, and a finite-sample interval valid across their lower-rank strata (Sections 4–5).
2. Two quantitative separations: from fixed-radius affine intervals under the same honesty class, and from all intervals under a single-unfolding relaxation (Section 5).
3. An explicitly attributed classical diagonal-information benchmark and a full-data scan attaining the optimal expected-length rate at zero for $K \geq 4$ (Sections 5.4–5.5).
4. A harmonic full-data lower bound for $K = 3$, leaving a logarithmic gap at zero. The simulations evaluate the original threshold interval; the scan is an existence result (Sections 5.5–6).

2 Related work

Gaussian functionals and approximate sparsity. Optimal recovery and nonconvex functional estimation connect risk to the geometry of the parameter set [Donoho, 1994, Cai and Low, 2004b]. In the Gaussian sequence model $y = \theta + \sigma g$, Collier et al. [2017] give sharp rates for $\sum_i \theta_i$ over ℓ_q balls. In their sparse regime, the squared-error scale is

$$\sigma^2(r/\sigma)^{2q}\{\log[1 + d(\sigma/r)^{2q}]\}^{1-q}, \qquad \|\theta\|_q \leq r, \quad 0 < q \leq 1.$$

The specialization $q = 2/K$, $r = M = \tau\sqrt{p}$, $d = p$ and division by p for our normalized contrast yields the squared diagonal-information rate below. We explicitly credit this benchmark and

give its confidence-interval derivation. Its transfer to the full tensor experiment would require a different lower bound. Our full-data scan and harmonic mixture provide separate arguments in that experiment.

Honest adaptation and weak ranks. The confidence-interval theory of Cai and Low [2004a] uses between-class moduli and testing to characterize adaptation, while Low [1997] gives foundational limitations. Finite convex-union aggregation is developed by Juditsky and Nemirovski [2020]. Thus, neither nonlinear inference nor failure of uniform adaptation at a model boundary is peculiar to tensors. Matrix confidence sets and detection results further show that attainable rank adaptation depends on the observation design and loss [Carpentier et al., 2018, Carpentier and Nickl, 2015].

In econometric models, Choi et al. [2026] study $Y_{it} = X_{it}M_{it} + E_{it}$ and construct studentized block-average inference with a projection dimension at least the true rank. Their quantitative strength and alignment assumptions remain relevant even though exact rank is not estimated. Armstrong et al. [2026] consider a regression coefficient in a panel with a low-rank interactive effect, using weights and a bias bound for uniform inference without a minimum factor strength. Zhu [2019] establishes weak-factor and rank-adaptation obstructions in a missing-entry experiment. These are different targets and designs from the fully observed tensor contrast considered here.

Regular matrix and tensor functional inference. For tensor regression $Y_i = \langle X_i, A\rangle + \epsilon_i$, Xu et al. [2025] combine a pilot estimator with debiasing and tangent projection to infer general linear functionals. Their regular-stratum results require rank/subspace control and signal/remainder conditions; the principal regression theorem imposes a positive minimum unfolding signal. Its tangent standard deviation is of the form $\sigma\|P_{T_A}L\|_F/\sqrt{n}$. Our replacement is uniform finite-sample coverage on a cap union, at the cost of a narrower loading, known norm/noise bounds and potentially conservative length. No pointwise dominance over that efficient regime is asserted.

Related matrix linear-form inference, geometric inverse-problem inference and limiting distributions under rank caps are developed by Xia and Yuan [2021], Cai et al. [2016], Carpentier and Kim [2018]. Tensor subspace and functional inference includes strong-signal results of Xia et al. [2022], Agterberg and Zhang [2024]; Cai et al. [2023] treats tensor completion and entry/factor uncertainty. Covariance-weighted low-rank regression inference [Ke et al., 2026] and uniform inference for weak matrix signals [Bykhovskaya et al., 2025] provide recent neighboring results. Their observation models and regularity conditions cannot be replaced by the Gaussian location experiment without proof.

Ma et al. [2026] also give a tensor-completion confidence-length lower bound in their Theorem 10. Its class imposes incoherence, a positive minimum unfolding signal, a condition-number cap and nonvanishing tangent-loading alignment. It excludes zero and uses randomly sampled entries; it does not supply a lower bound for the rank-boundary Gaussian location problem here.

Geometry and finite-sample inference. Orthonormal Tucker and TT representations are standard [De Lathauwer et al., 2000, Oseledets, 2011]. Tensor norms and Euclidean factor penalties already contain the exponent $2/K$ [Friedland and Lim, 2018, Fan et al., 2023]; tangent cones of Tucker and TT varieties have also been studied [Gao et al., 2025, Kutschan, 2018]. We use canonical expansions without estimating tangent spaces. Singular-model likelihood theory [Drton, 2009] and universal inference [Wasserman et al., 2020] establish broader approaches to nonregular validity. Our specific comparison concerns interval length under simultaneous rank constraints, rather than generic existence of a valid interval.

The comparison can be written as

$$\text{regular-stratum functional inference} \longrightarrow \text{cap-uniform inference for a restricted loading,}$$

and, within the present Gaussian experiment,

$$\text{honesty on one unfolding class} \longrightarrow \text{honesty on the joint tensor class.}$$

The first changes assumptions and loading scope; the second permits a quantitative length separation. The companion survey records 20 closest papers and their mathematical differences, including relevant work available by September 9, 2026.

3 Experiment and parameter classes

Let $K \geq 2$, $p_k \geq 1$, and $D = \prod_{k=1}^{K} p_k$. We observe

$$Z = A + \tau G \in \mathbb{R}^{p_1 \times \cdots \times p_K}, \qquad \operatorname{vec}(G) \sim N(0, I_D), \qquad \tau > 0. \tag{1}$$

The noise level is known. Equivalently, Z is the sufficient sample mean of independent full observations $W_j = A + \sigma G_j$, $j = 1, \ldots, n$, with known σ and $\tau = \sigma/\sqrt{n}$. This is an exact Gaussian mean equivalence, not an equivalence to scalar-response tensor regression. The observed variables are the tensors W_j, or Z; A is a fixed unknown parameter. Factor and core representations are nonunique coordinates, not additional latent random variables.

Write $A_{(k)}$ for the mode-k unfolding and $A_{\langle k \rangle}$ for the matrix joining modes $1, \ldots, k$ against $k+1, \ldots, K$. For supplied positive integer caps, define

$$\Theta_{\mathrm{Tu}}(M, \bar{r}) = \{A : \|A\|_F \leq M,\ \operatorname{rank}(A_{(k)}) \leq \bar{r}_k,\ k = 1, \ldots, K\}, \tag{2}$$

$$\Theta_{\mathrm{TT}}(M, \bar{r}) = \{A : \|A\|_F \leq M,\ \operatorname{rank}(A_{\langle k \rangle}) \leq \bar{r}_k,\ k = 1, \ldots, K-1\}. \tag{3}$$

The known bound M is positive. Actual ranks may be smaller, including zero; nonzero singular values may be arbitrarily small. No incoherence or coordinate sparsity condition is imposed. The main experiment assumes Gaussian isotropic noise and exact caps. Appendix C states separately calibrated noise and approximation extensions; missing observations remain untreated.

The prescribed loading has the form

$$L = \sum_{i=1}^{m} w_i V_i, \qquad V_i = \bigotimes_{k=1}^{K} v_{ki}, \qquad v_{ki}^{\top} v_{kj} = \mathbf{1}\{i = j\}, \qquad \|w\|_2 = 1. \tag{4}$$

Here $m \leq \min_k p_k$, and each loading family is supplied independently of the data. Thus $\|L\|_F = 1$. Although this is restrictive, L need not be sparse in the observation basis: known orthogonal mode rotations can make it dense. The estimand and relevant observations are

$$\psi(A) = \langle L, A \rangle = \sum_i w_i a_i, \qquad a_i = \langle A, V_i \rangle, \qquad d_i = \langle Z, V_i \rangle = a_i + \tau g_i, \tag{5}$$

where the g_i are independent standard normals.

An interval $C(Z)$ is honest at level $1 - \alpha$ on Θ if

$$\inf_{A \in \Theta} \mathbb{P}_A\{\psi(A) \in C(Z)\} \geq 1 - \alpha. \tag{6}$$

This is a finite-sample requirement, not a pointwise limit along a fixed-rank stratum. Length and expected length are the primary criteria. For example,

$$\mathcal{L}_0(\Theta) = \inf_{C:\,(6)} \mathbb{E}_0|C|, \qquad \mathcal{L}_{\max}(\Theta) = \inf_{C:\,(6)} \sup_{A\in\Theta} \mathbb{E}_A|C|. \tag{7}$$

The infima range over all measurable intervals, allowing independent randomization. The usual squared Frobenius estimation loss has population risk $R_A(B) = \|B - A\|_F^2 + D\sigma^2$ and empirical risk $R_n(B) = n^{-1}\sum_j \|W_j - B\|_F^2$. We do not minimize these risks or require a full-tensor estimator.

The principal asymptotic comparison in experiment (1) uses $p_k = m = p$, fixed $K \geq 3$, fixed rank caps, and $M = \tau\sqrt{p}$. A concrete sequence is $M = \sigma = 1$, $n = p$, so $D = n^K$. All finite-sample results below also apply away from that sequence.

The structural and probabilistic assumptions have distinct roles, summarized in Table 1. A1–A4 define the inferential problem; A5 is used only for the asymptotic comparisons.

	Assumption	Role and restriction
A1	Full observation with independent Gaussian coordinates and known τ	Gives exact diagonal noise, direct calibration and Gaussian likelihood identities.
A2	Supplied M and Tucker or TT upper rank bounds	Defines the outer honesty class; actual rank and signal strength are unrestricted below it.
A3	Prescribed loading in (4)	Gives orthogonal noise coordinates and the diagonal envelope; data-selected or arbitrary loadings are excluded.
A4	Finite-sample uniform coverage in (6)	Coverage includes zero and local sequences approaching lower-rank strata.
A5	Equal dimensions, fixed $K \geq 3$, fixed caps and $M = \tau\sqrt{p}$	Produces the stated vanishing length ratio; not needed for the finite-sample interval.

Table 1: Assumptions, their purpose and the scope they exclude. Known upper bounds are not replaced by same-data estimates anywhere in the stated procedure.

4 Interval construction

Define

$$q = \frac{2}{K}, \quad J_{\mathrm{Tu}} = \min_k \prod_{\ell\neq k} \bar{r}_\ell, \quad J_{\mathrm{TT}} = \prod_{k=1}^{K-1} \bar{r}_k, \quad B = J^{1-q/2}M^q, \tag{8}$$

where J is chosen for the applicable class. Set

$$\lambda = \tau\sqrt{2\log(2m/\alpha)}, \qquad \widehat{\psi}_{\mathrm{NL}} = \sum_i w_i \,\mathrm{soft}_\lambda(d_i), \qquad \mathrm{soft}_\lambda(x) = \mathrm{sign}(x)(|x| - \lambda)_+. \tag{9}$$

The three candidate intervals have radii

$$h_{\mathrm{NL}} = \|w\|_\infty B(2\lambda)^{1-q}, \qquad I_{\mathrm{NL}} = [\widehat{\psi}_{\mathrm{NL}} - h_{\mathrm{NL}}, \widehat{\psi}_{\mathrm{NL}} + h_{\mathrm{NL}}], \tag{10}$$

$$h_G = z_{1-\alpha/2}\tau, \qquad I_G = [\langle L, Z\rangle - h_G, \langle L, Z\rangle + h_G], \tag{11}$$

$$h_0 = M\left(\sum_{i=1}^{s} w_{(i)}^2\right)^{1/2}, \qquad I_0 = [-h_0, h_0], \tag{12}$$

where $|w_{(1)}| \geq \cdots \geq |w_{(m)}|$ and s is the smaller of m and the smallest supplied rank cap. Choose

$$C_* = I_{j_*}, \qquad j_* \in \arg\min_{j \in \{\mathrm{NL}, G, 0\}} h_j, \tag{13}$$

with a fixed rule for ties. This choice depends only on known quantities. Choosing an interval after examining its center or an estimated rank is a different procedure and is not justified by the proof.

The constants in (8) calibrate the threshold procedure (9); the direct alternative (11) supplies a valid fallback. Only m contractions are used. For coordinate loadings, computation reads m tensor entries and takes $O(m)$ operations. Obtaining contractions for a dense rotated loading has additional cost, which can be large. The procedure does not estimate a Tucker core, TT cores, factors or actual ranks.

5 Statistical theory

All numbered theorems and propositions are **PROVED**, with complete derivations in the appendix. The result-status table there distinguishes these claims from **PARTIAL** and **CONJECTURED** extensions. The results use the experiment and classes of Section 3. Lower-bound constants are stated for $0 < \alpha < 1/4$ when this simplifies positivity.

5.1 Diagonal geometry and honest coverage

Theorem 1 (Orthogonal diagonal compressibility)**.** *For either class in* (2)–(3)*, under loading* (4)*,*

$$\sum_{i=1}^{m} |\langle A, V_i \rangle|^{2/K} \leq J^{1-1/K} M^{2/K}. \tag{14}$$

No minimum singular value is required.

The proof expands a canonical tensor representation into at most J product terms whose squared component weights sum to $\|A\|_F^2$. For TT tensors, these terms need not be orthogonal. A prefix-weight identity establishes the sum exactly; assuming termwise orthogonality would be an incorrect shortcut. Hölder's inequality and Bessel's inequality then give (14). The rank factor may be loose, particularly for growing caps.

For rank cap one the diagonal image is exactly the ball $\{a : \sum_i |a_i|^{2/K} \leq M^{2/K}\}$, as shown in Lemma 2. Thus the diagonal-only statistical experiment is precisely a classical Gaussian sequence problem on an ℓ_q ball. The full tensor experiment contains additional off-diagonal information. This identity permits a sharp benchmark for the restricted observation experiment; it does not remove the additional information in the rest of the tensor.

Theorem 2 (Finite-sample honest interval)**.** *For every $0 < \alpha < 1$, each interval in* (10)–(12) *is honest on the applicable cap class. Consequently, C_* is honest and has deterministic length*

$$|C_*| = 2 \min\{h_{\mathrm{NL}}, h_G, h_0\}. \tag{15}$$

The event $\max_i |\tau g_i| \leq \lambda$ has probability at least $1 - \alpha$. On this event soft thresholding satisfies

$$|\operatorname{soft}_\lambda(a_i + \tau g_i) - a_i| \leq \min(|a_i|, 2\lambda) \leq |a_i|^q (2\lambda)^{1-q}.$$

Summing proves the nonlinear interval bound. No central limit theorem, rank recovery or pilot remainder is needed. Known M is essential to this deterministic calibration.

5.2 Two lower-bound comparisons

For the remainder of this subsection let all dimensions equal p, let $E_i = e_i^{\otimes K}$, and put

$$L = p^{-1/2} \sum_{i=1}^{p} E_i, \qquad z = z_{1-\alpha/2}. \tag{16}$$

Theorem 3 (Fixed-radius affine intervals)**.** *Suppose the honesty class contains* 0 *and all* $\pm ME_i$*. If* $0 < \alpha < 1/2$ *and*

$$C(Z) = [\langle W, Z\rangle + b - h, \langle W, Z\rangle + b + h]$$

is honest, with deterministic W, b, h*, then*

$$h \geq \frac{(M/\sqrt{p})z\tau}{M/\sqrt{p} + z\tau}. \tag{17}$$

On the rank-one tensor Frobenius ball, the best radius in this affine class is therefore of order $\min(M/\sqrt{p}, \tau)$*, up to constants depending on* α*.*

The theorem allows W to use the entire tensor. It does not cover affine centers with data-dependent radii. The matching upper order follows by choosing between I_G and I_0 for rank cap one.

For a nonempty proper subset S of modes, write $A_{[S]}$ for the corresponding bipartition unfolding and define

$$\Theta_S = \{A : \operatorname{rank}(A_{[S]}) \leq 1,\ \|A\|_F \leq M\}. \tag{18}$$

Theorem 4 (Cost of a single unfolding honesty class)**.** *If* $0 < \alpha < 1/4$*, every interval honest on* Θ_S *satisfies*

$$\mathbb{E}_0|C| \geq c_\alpha \min\left\{\frac{M}{\sqrt{p}}, \frac{\tau}{4}\right\}, \qquad c_\alpha = 1 - 2\alpha - \frac{1}{2}\sqrt{\sqrt{8/7} - 1} > 0. \tag{19}$$

The lower order $\min(M/\sqrt{p}, \tau)$ *is attained up to* α*-dependent constants by the direct/structural choice on* Θ_S*.*

The alternatives in the proof are rank one in the chosen matrix unfolding. They generally violate the other tensor rank constraints. This is necessary for the comparison: the theorem measures the cost of requiring coverage on the larger class. For Tucker caps, a mode unfolding is a permissible relaxation; for TT caps, a contiguous cut is permissible. With all tensor rank caps one, every bipartition in (18) contains the original tensor class. Increasing an unfolding cap further preserves the lower bound.

Theorem 5 (Strict length separations)**.** *Let* $0 < \alpha < 1/4$*,* $K \geq 3$ *and the rank caps be fixed. Under* (16) *and* $M = \tau\sqrt{p}$*,* C_* *is honest on the Tucker or TT class and*

$$|C_*| \leq 2\tau J^{1-1/K} \left\{\frac{8\log(2p/\alpha)}{p}\right\}^{(K-2)/(2K)} = o(\tau). \tag{20}$$

In contrast, every fixed-radius affine interval honest on the same class has length at least $2z\tau/(1+z)$*. Every interval honest on a class retaining only one constrained unfolding has expected length at zero at least* $c_\alpha\tau/4$*.*

For growing caps, the same conclusion follows if $J^{1-1/K}(\log p/p)^{(K-2)/(2K)} \to 0$. This condition may be restrictive. At $K = 2$ the exponent in (20) is zero, so the argument gives no polynomial separation. Theorem 5 neither asserts that the displayed exponent is optimal nor compares favorably at every strong signal with an efficient oracle projection interval. Figure 1 displays the explicit comparisons and the within-class bracket established next.

5.3 An initial full-data lower bound

Theorem 6 (Within-class lower bound)**.** *Under* (1) *and* (16)*, for* $0 < \alpha < 1/4$*, every interval honest on a class containing the rank-one tensor Frobenius ball of radius* M *satisfies*

$$\mathbb{E}_0|C| \geq (3/4 - 2\alpha)\frac{\min\{M, \tau\sqrt{\log(1 + p/4)}\}}{\sqrt{p}}. \tag{21}$$

This is a sparse-coordinate mixture bound. At $M = \tau\sqrt{p}$, for fixed caps and fixed $K \geq 3$, the results imply

$$c\tau\sqrt{\frac{\log p}{p}} \leq \mathcal{L}_0(\Theta) \leq \mathcal{L}_{\max}(\Theta) \leq C\tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)} \tag{22}$$

for sufficiently large p, with constants depending on K, the caps and α. The ratio of these initial scales grows as $(p/\log p)^{1/K}$. Section 5.5 improves the full-data zero-point analysis: it closes that gap for $K \geq 4$ and reduces it to a logarithmic factor for $K = 3$. The worst-case criterion remains unmatched, and the two criteria in (7) need not agree.

5.4 Optimality when only the diagonal is observed

Let Θ_1 denote the Frobenius ball of tensors with rank at most one, including zero. Restrict the interval to be measurable in $d = (Z_{i,\ldots,i})_{i=1}^p$ and an independent random seed. Write $\mathcal{C}_{\alpha,d}(\Theta_1)$ for such intervals that are honest on Θ_1, and define

$$\mathcal{L}_{0,d}(\Theta_1) = \inf_{C \in \mathcal{C}_{\alpha,d}(\Theta_1)} \mathbb{E}_0|C|, \qquad \mathcal{L}_{\max,d}(\Theta_1) = \inf_{C \in \mathcal{C}_{\alpha,d}(\Theta_1)} \sup_{A \in \Theta_1} \mathbb{E}_A|C|.$$

The next result specializes classical sparse-sequence arguments [Cai and Low, 2004b, Collier et al., 2017] to honest intervals. It is a benchmark for the proposed use of diagonal information, not a new Gaussian-sequence minimax principle.

Proposition 1 (Classical diagonal-information benchmark)**.** *Assume* (1)*,* (16) *and* $0 < \alpha < 1/4$*. For each integer* $1 \leq s \leq p$*, put*

$$b_s = \min\left\{Ms^{-K/2},\ \tau\sqrt{\log\left(1 + \frac{p}{4s^2}\right)}\right\}, \qquad c_{\alpha,d} = 1 - 2\alpha - \frac{1}{2}\sqrt{e^{1/4} - 1} > 0.$$

Every $C \in \mathcal{C}_{\alpha,d}(\Theta_1)$ *satisfies*

$$\mathbb{E}_0|C| \geq c_{\alpha,d}\frac{sb_s}{\sqrt{p}}. \tag{23}$$

For fixed $K \geq 3$ *and* $M = \tau\sqrt{p}$*,*

$$\mathcal{L}_{0,d}(\Theta_1) \asymp \mathcal{L}_{\max,d}(\Theta_1) \asymp \tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)}. \tag{24}$$

The comparison constants depend on K *and* α*.*

The rate in (24) follows because the effective support size $s \asymp (p/\log p)^{1/K}$ makes (23) match the threshold interval's upper order. The same asymptotic conclusion holds for any fixed Tucker or TT caps: the class contains Θ_1 and Theorem 2 changes the upper bound only by a fixed rank factor. Thus further improvement in the power of p cannot come from recalibrating a procedure restricted to d.

This result does not close the full-data gap in (22). For the common-target tensors $A_S = b_s(\sum_{i\in S} e_i)^{\otimes K}$, two support sets with overlap J have diagonal inner product $b_s^2 J$ but full-tensor inner product $b_s^2 J^K$. The likelihood second moment in the full experiment is consequently different. A valid diagonal lower bound cannot be transferred by ignoring the extra observations. Theorem 7 establishes that they do permit shorter expected length at zero.

Table 2 separates the procedure and parameter classes in the comparisons; Figure 1 displays the corresponding bounds.

Comparison	Information and honesty class	Length statement when $M = \tau\sqrt{p}$
Direct Gaussian interval	All means; prescribed L	$2z\tau$, with exact coverage.
Selected interval	Joint cap class; uses d	At most $2\tau J^{1-1/K}\{8\log(2p/\alpha)/p\}^{\beta}$; $\beta = (K-2)/(2K)$.
Any fixed-radius affine interval	Full tensor; same joint class	Length at least $2z\tau/(1+z)$.
Any interval on one unfolding class	Full tensor; enlarged honesty class	Expected length at zero at least $c_\alpha\tau/4$.
Any diagonal-based interval	Only d; rank-one or fixed-cap class	Optimal expected length at zero and in worst case has order $\tau(\log p/p)^{\beta}$.
Any full-data interval	Full tensor; joint class	At zero: sharp rate $\tau\sqrt{\log p/p}$ for $K \geq 4$; cubic bracket (31). Worst-case optimality remains open.

Table 2: Precisely delimited comparisons for fixed $K \geq 3$ and fixed rank caps. Lower bounds for relaxed honesty classes or restricted observations are not lower bounds for unrestricted full-data procedures on the original class.

5.5 Using the full tensor at the zero mean

The diagonal rate is a constraint on procedures using the diagonal. A finite scan over product directions of coordinate subtensors can give shorter expected length at zero while preserving the original uniform length bound. The scan is an existence construction with exponential complexity; no efficient implementation is asserted.

For a tensor B, define its injective norm

$$\rho(B) = \sup_{\|u_k\|_2=1} |\langle B, u_1 \otimes \cdots \otimes u_K\rangle|.$$

For every nonempty subset $S \subseteq \{1,\ldots,p\}$ of size s, choose fixed $1/(2K)$-nets of the unit sphere in $\mathbb{R}^s$, one per mode, each with at most $(1+4K)^s$ points. Let $N_S(Z)$ be the largest absolute contraction of $Z_{S,\ldots,S}$ over their product. Define

$$\lambda_s^{\mathrm{scan}} = \tau\left[2\{s\log(ep/s) + Ks\log(1+4K) + \log(2p/\gamma)\}\right]^{1/2}, \tag{25}$$

$$R_\gamma = \frac{4J^{K/2}}{\sqrt{p}} \sum_{s=1}^{p} \lambda_s^{\mathrm{scan}} s^{-K/2}. \tag{26}$$

Accept the zero test if $N_S(Z) \leq \lambda_{|S|}^{\mathrm{scan}}$ for every nonempty S. On acceptance, let $C_{\mathrm{scan}} = [-h_{\mathrm{scan}}, h_{\mathrm{scan}}]$, where $h_{\mathrm{scan}} = \min(h_0, R_\gamma)$; on rejection let $C_{\mathrm{scan}} = [-h_0, h_0]$. Here $h_0 = M\sqrt{s_{\mathrm{cap}}/p}$ and s_{cap} is the smallest cap, truncated at p. For $0 < \alpha_0, \gamma < 1$ with $\alpha_0 + \gamma \leq \alpha$, intersect this interval with $C_*(\alpha_0)$ and call the result $C_\dagger$; replace an empty intersection by $\{0\}$. Write $\ell_* = |C_*(\alpha_0)|$.

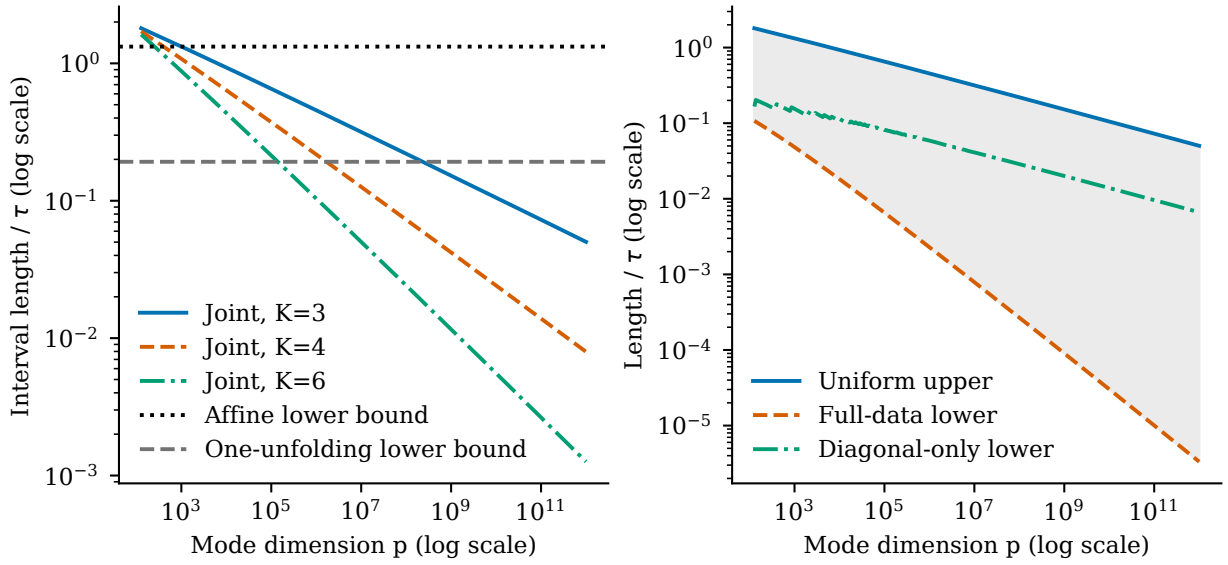


Figure 1: Joint constraints permit vanishing normalized length, and the diagonal-only lower bound matches its order. Left: cap-one selected length and the affine and one-unfolding lower bounds, for $M = \tau\sqrt{p}$ and $\alpha = 0.05$. Right: the $K = 3$ upper bound, the full-data lower bound and the diagonal-only lower bound from (23), using $s = \lfloor (p/\log p)^{1/3} \rfloor$. The right panel shows the initial bounds, before the improvements in Theorems 7–8; it is not the final full-data bracket. Curves evaluate analytic formulas up to $p = 10^{12}$; simulated dimensions stop at 8192.

Theorem 7 (A full-data interval with short expected length at zero)**.** *Under* (1), (16) *and either cap class,* C_{scan} *is honest at level* $1-\gamma$*, and* $C_\dagger$ *is honest at level* $1-\alpha$*. Moreover,*

$$\sup_{A\in\Theta} \mathbb{E}_A|C_\dagger| \le \ell_*, \qquad \mathbb{E}_0|C_\dagger| \le \min\{\ell_*,\, 2R_\gamma + \ell_*\gamma\}. \tag{27}$$

For fixed $0 < \alpha < 1/4$*, fixed* $K \ge 3$*, fixed positive caps and* $M = \tau\sqrt{p}$*, choose* $\alpha_0 = \alpha/2$ *and* $\gamma = \alpha/(2p)$*. Then*

$$\mathbb{E}_0|C_\dagger| \le C_{K,J,\alpha} \frac{\tau\sqrt{\log p}}{\sqrt{p}} \sum_{s=1}^{p} s^{-(K-1)/2} \tag{28}$$

for sufficiently large p*. Consequently,*

$$\mathcal{L}_0(\Theta) \asymp \tau\sqrt{\frac{\log p}{p}}, \qquad K \ge 4. \tag{29}$$

The matching lower bound in (29) is Theorem 6, applied to the same full-data experiment and honesty class. The sum in (28) is bounded for $K \ge 4$ and harmonic for $K = 3$. The number of scanned product directions is bounded by $(1 + (1 + 4K)^K)^p - 1$. Thus the result establishes attainable statistical precision, without a polynomial-time guarantee. The worst-case bound in (27) preserves the original diagonal upper order and remains unmatched by a full-data worst-case lower bound.

Theorem 8 (A harmonic obstruction in order three)**.** *Under* (1) *and* (16)*, let* $K = 3$*,* $M = \tau\sqrt{p}$ *and* $0 < \alpha < 1/4$*. Every interval honest on a class containing the rank-one tensor Frobenius ball satisfies*

$$\mathbb{E}_0|C| \geq (3/4 - 2\alpha)\frac{\tau H_p}{200\sqrt{p}}, \qquad H_p = \sum_{i=1}^{p} i^{-1}. \tag{30}$$

Theorem 8 uses a full-data mixture of rank-one tensors with sign-coupled factors and a common harmonic diagonal. It refutes the earlier conjecture that $\mathcal{L}_0$ always has order $\tau\sqrt{\log p/p}$ at rank one. Combining (30) and (28) gives the revised cubic bracket

$$c_\alpha^{\mathrm{cub}}\frac{\tau \log p}{\sqrt{p}} \leq \mathcal{L}_0(\Theta) \leq C_{J,\alpha}\frac{\tau(\log p)^{3/2}}{\sqrt{p}}. \tag{31}$$

Here $c_\alpha^{\mathrm{cub}} = (3/4 - 2\alpha)/200 > 0$. Only a factor $\sqrt{\log p}$ remains in (31). The constant $1/200$ is conservative and is not an optimal finite-sample constant.

Further verified refinements. The accompanying theory review proves a smaller rank coefficient in the diagonal bound, an exact one-dimensional characterization of the optimal fixed affine radius, and a noise-envelope version with an additive approximation term. It also records finite-variance and unknown-Gaussian-variance extensions requiring retained replicates. These supporting results are collected in Appendix C; the simulations retain the original Gaussian threshold procedure.

Assumption audit. Gaussian isotropy gives the exact diagonal sequence and the likelihood-ratio identities. The threshold interval alone extends whenever a valid simultaneous coordinate noise envelope is supplied, but the direct alternative and lower bounds require their own calibration. Orthonormal loading families produce Bessel bounds and equal noise variances; arbitrary loadings are untreated. Canonical tensor expansions avoid a spectral gap but yield potentially large rank constants. Supplied caps and M identify the outer honesty class; replacing them with unvalidated estimates is not covered. None of these assumptions should be described as a routine approximation for dependent LLM representations.

6 Simulation study

6.1 Protocol and baselines

The simulation protocol was written after the theorem derivations and before implementation. It uses seed 20260909, 2,000 replications per configuration, and exact marginal observations of the diagonal coordinates needed by the procedures. Unused independent Gaussian coordinates of the full tensor need not be generated. This does not simulate a regression design or a tensor completion experiment.

The main grid takes $K \in \{2, 3, 4, 6\}$, $p = n \in \{128, 512, 2048, 8192\}$, $M = \sigma = 1$, and rank cap one. Signals have the form $A = au^{\otimes K}$, with $a \in \{0, 0.02, 0.2, 1\}$ and u uniform on a coordinate support. For nonzero signals, support sizes include one, p, and a value near $(a/\lambda)^{2/K}$, with duplicates removed. The latter places diagonal means near the threshold. Additional configurations vary n/p and the valid threshold multiplier, and use cap two with a second component tending to zero. There are 175 configurations, of which three deliberately violate the iid Gaussian model.

We compare I_G, I_0, I_{NL} and C_*. For nonzero rank-one signals, a known-direction oracle is also included. If $U = u^{\otimes K}$ were supplied, then $c\langle U, Z\rangle \pm z\tau|c|$, where $c = \langle L, U\rangle$, is exact for $A = aU$. Its

joint Gaussian noise with the diagonal is simulated exactly. This oracle has unavailable structural information and a smaller honesty class; it is not an implementation of an estimated-subspace method. Published strong-signal methods are compared through their assumptions rather than advertised as uniform baselines without a matching implementation.

Each output row reports coverage, binomial Monte Carlo standard error, an exact 95% Clopper–Pearson interval, deterministic length, center bias and its standard error, and root mean squared center error. Width standard errors are zero because the radii are deterministic. Numerical proof checks separately examine 101,101 scalar soft-threshold cases, 120 random Tucker and 120 random TT tensors, and exact finite-binomial mixture second moments. The diagonal benchmark is additionally checked against an enumeration of 3,136 support pairs and exact hypergeometric moments in 24 configurations. They supplement rather than replace the proofs.

6.2 Gaussian coverage and interval length

Table 3 summarizes the 172 in-model configurations. The selected interval is substantially conservative. Across configurations, direct Gaussian coverage ranges from 0.9365 to 0.9625, and selected coverage ranges from 0.9975 to 1. These extrema are descriptive, not simultaneous confidence limits for uniform coverage. At coverage 0.95, the Monte Carlo standard error with 2,000 replications is approximately 0.00487. When all replications cover, the exact binomial interval is approximately $[0.99816, 1]$, despite a zero plug-in standard error.

Method	Configurations	Min. coverage	Max. coverage
Direct Gaussian	172	0.9365	0.9625
Structural	172	1.0000	1.0000
Threshold	172	0.9975	1.0000
Selected	172	0.9975	1.0000
Known-direction oracle	117	0.9370	0.9615

Table 3: Coverage extrema over the stated Gaussian configurations. The oracle is evaluated only for nonzero rank-one signals, giving fewer configurations and a different information set. Each configuration has 2,000 replications; configuration-level uncertainty is retained in the accompanying CSV.

At $p = n = 8192$ and cap one, the selected lengths divided by τ are 2.0000, 0.9622, 0.6674 and 0.4629 for $K = 2, 3, 4, 6$, respectively. The direct length is 3.9199τ. These width values are exact evaluations of the formulas, not fitted empirical convergence rates. They agree with the order-dependent prediction, while the $K = 2$ control shows no vanishing length ratio. A known-direction oracle can be much shorter for diffuse signals; there is no claim of pointwise oracle dominance.

Table 4 examines center accuracy for three support sizes in one fixed configuration. Near the threshold, the selected center has visible shrinkage bias. Its deterministic error envelope covers that bias, which helps explain the conservative coverage. At a diffuse signal the known-direction oracle uses information that the selected interval does not exploit.

6.3 Weak rank and noise assumptions

For the boundary experiment, set $K = 3$ and cap two, and generate

$$A_\epsilon = \sqrt{1 - \epsilon^2}\, U_1 + \epsilon U_2, \qquad \epsilon \in \{0, 0.001, 0.01, 0.1, 0.5\},$$

Support	Procedure	Bias (MCSE)	RMSE	Length	Coverage
1	Selected	-0.0555 (0.00025)	0.0566	0.962	1.0000
1	Direct	0.01 (0.0223)	0.997	3.92	0.9450
1	Oracle	0.000204 (0.00025)	0.0112	0.0433	0.9465
7	Selected	-0.352 (0.000346)	0.353	0.962	1.0000
7	Direct	-0.0272 (0.022)	0.985	3.92	0.9575
7	Oracle	0.000235 (9.21×10^{-5})	0.00412	0.0164	0.9550
8192	Selected	-0.011 (4.36×10^{-6})	0.011	0.962	1.0000
8192	Direct	0.00976 (0.0229)	1.02	3.92	0.9445
8192	Oracle	5.78×10^{-6} (2.74×10^{-6})	0.000122	0.000479	0.9480

Table 4: Center accuracy and length at $K = 3$, $p = n = 8192$, $a = M = \sigma = 1$ and cap one. Bias, its Monte Carlo standard error in parentheses, RMSE and length are divided by τ. Each row uses 2,000 replications. The support sizes represent concentrated, near-threshold and diffuse signals; the complete grid is retained in the results.

where U_1, U_2 are unit product tensors with disjoint mode supports. Each constrained unfolding has nonzero singular values $\sqrt{1-\epsilon^2}$ and ϵ for $\epsilon > 0$; the actual rank becomes one at zero. Figure 2 shows coverage and length for concentrated supports at $p = 8192$. The selected radius remains constant as ϵ vanishes. This demonstrates validity across the tested boundary and explicitly demonstrates the absence of actual-rank adaptation.

Three additional zero-mean stress configurations use independent variance-one Student t_3 errors, a Gaussian mixture with 5% of noise standard deviations multiplied by ten, and equicorrelated Gaussian coordinates with correlation 0.2. Table 5 reports the results. Variance contamination causes severe undercoverage for both procedures. Correlated Gaussian coordinates invalidate the direct iid variance but preserve the marginal Gaussian union bound used for the nonlinear interval. The Student result at zero is not a robustness theorem or evidence of coverage elsewhere.

Noise	Direct	Selected
gaussian	0.9540	1.0000
student t3	0.9585	0.9670
variance contamination	0.5740	0.3830
correlated	0.1430	1.0000

Table 5: Noise-assumption stress at $A = 0$, $K = 3$, $p = n = 512$, cap one and nominal level 0.95. Only the first row belongs to the stated iid Gaussian model. These results delimit the calibration; they do not support a robustness claim.

7 Discussion

The results show that the parameter class over which coverage is required can determine the order of an honest interval's length. For a prescribed orthogonal contrast, simultaneous tensor rank restrictions yield an interval smaller than the order required by a single-unfolding relaxation. The comparison holds at arbitrarily weak signals and does not depend on identifying the actual rank. It also separates nonlinear procedures from every fixed-radius affine interval on the original class.

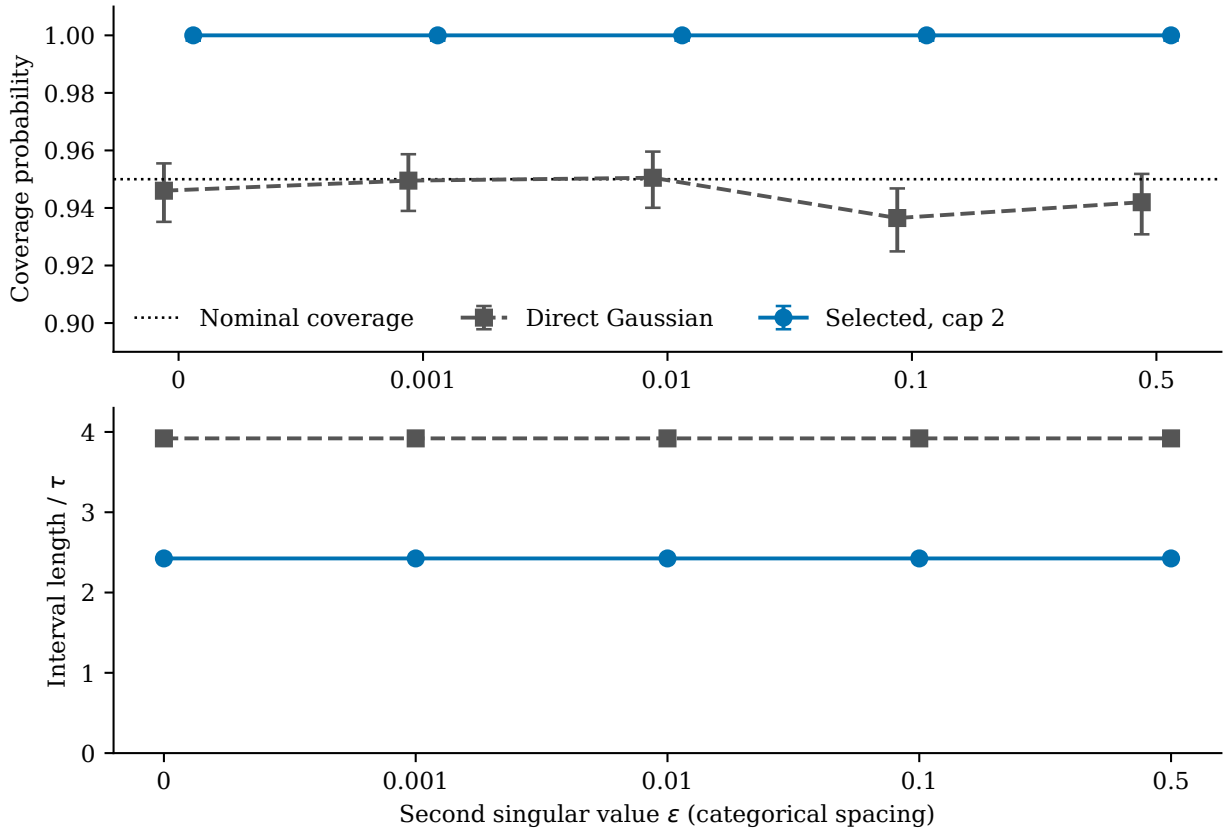


Figure 2: Coverage persists in the tested weak-component sequence, but the cap-based radius does not adapt to the actual rank. The setting is $K = 3$, $p = n = 8192$, $M = \sigma = 1$ and cap two, with disjoint single-coordinate product factors. Top: empirical coverage and exact 95% binomial intervals from 2,000 replications per value. Bottom: deterministic length divided by τ. The horizontal spacing in ϵ is categorical.

The exact diagonal reduction gives a precise connection with classical sequence inference. At rank one, the attainable diagonal vectors form an $\ell_{2/K}$ ball; Proposition 1 then identifies the optimal rate among diagonal-based intervals. The full-data scan quantifies the value of the off-diagonal observations: $\mathcal{L}_0 \asymp \tau\sqrt{\log p/p}$ for fixed $K \geq 4$, whereas the harmonic mixture gives $\mathcal{L}_0 \gtrsim \tau \log p/\sqrt{p}$ for $K = 3$. The remaining cubic gap is $\sqrt{\log p}$. None of these conclusions identifies the full-data worst-case optimum. Similarly, Theorem 4 applies to a larger honesty class, not to every algorithm that represents its data as a matrix.

Several restrictions affect practical inference. The loading must have the orthogonal product form, the Frobenius and rank caps are supplied, and the noise standard deviation is known. Appendix C improves the cap coefficient and relaxes the noise and exact-rank assumptions under explicit calibration. These refinements do not provide automatic adaptation when a singular value vanishes. The experiments show substantial conservatism even though length improves over the specified benchmarks. Strong-signal subspace inference or a known-direction oracle may give better pointwise precision.

Closing the logarithmic gap for $K = 3$ is the most direct next question. It requires a sharper scan or a more difficult full-tensor mixture. The scan also raises a computational question: its

exhaustive finite construction is exponential, and no polynomial-time procedure attaining its zero-point guarantee has been established here. A second question is loading geometry: characterize which contrasts permit a joint-constraint advantage and which inherit the single-unfolding rate. Honest adaptation to actual rank would additionally require valid coverage on the larger cap class and a characterization of testable smaller-rank submodels. Rank consistency at separated signals alone is insufficient.

Applications require a sampling model that justifies the noise calibration and an independently specified contrast. Repeated calibrated tensor measurements could provide such a model, but no real-data experiment is claimed here. In particular, arrays indexed by LLM model, layer and prompt generally have prompt, token and layer dependence, heterogeneous noise and data-derived loadings. Simply stacking those observations does not establish the Gaussian experiment. The noise-envelope extension in Appendix C is one ingredient, but a justified dependence model and functional-selection theory would still be necessary before an LLM experiment could substantiate these inferential claims.

Reproducibility. The accompanying project contains the simulation protocol, modular Python code, fixed seeds, 817 result rows, exact binomial uncertainty, source and result hashes, the literature matrix and a theorem-status ledger. The original simulation outputs are reused without changing the interval algorithm. The theory review adds exact finite checks of the new sign-mixture calculation, approximation inequalities and affine bounds. No simulation of the exhaustive scan or new empirical coverage result is claimed. All displayed numerical values come from saved outputs or analytic formulas.

A Complete proofs

The proofs apply in finite dimensions. Constants used in the main text are explicit. All tensors and Gaussian variables are real.

A.1 Canonical product expansions

Lemma 1 (Controlled squared component weights)**.** *For a tensor in either cap class there is a finite representation*

$$A = \sum_{j=1}^{J_A} x_{1j} \otimes \cdots \otimes x_{Kj}, \qquad a_j^{\mathrm{cp}} = \prod_{k=1}^{K} \|x_{kj}\|_2,$$

such that $J_A \le J$ *and* $\sum_j (a_j^{\mathrm{cp}})^2 = \|A\|_F^2$.

Roadmap. Choose orthonormal Tucker mode bases or left-orthogonal TT cores; expand product terms and sum their squared weights. The TT argument uses a prefix identity, not orthogonality of individual paths.

Proof. The zero tensor has a zero-term representation. For a nonzero Tucker tensor, choose orthonormal bases U_k for the column spaces of its mode unfoldings. Orthogonal projection onto each such space leaves A unchanged. Contracting these bases gives a core $\mathcal{G}$ with $A = \mathcal{G} \times_1 U_1 \cdots \times_K U_K$ and $\|\mathcal{G}\|_F = \|A\|_F$. The actual mode ranks satisfy $r_k \le \bar{r}_k$.

Fix a mode k. Expand all core indices other than k. Each tuple j_{-k} gives unit factors $U_\ell[:, j_\ell]$ in modes $\ell \ne k$ and factor $U_k \mathcal{G}[:, j_{-k}]$ in mode k. Its squared product weight is $\|\mathcal{G}[:, j_{-k}]\|_2^2$. These slice squares sum to $\|\mathcal{G}\|_F^2$. The number of terms is $\prod_{\ell \ne k} r_\ell$. Choosing k to minimize the corresponding cap product proves the Tucker assertion.

For a TT tensor, sequential matrix factorizations yield cores $G_k \in \mathbb{R}^{r_{k-1}\times p_k\times r_k}$, with $r_0 = r_K = 1$ and $r_k \le \bar{r}_k$. At each cut, factor the relevant matrix through a basis of its column space. Equivalently, left-orthogonalize the cores by QR and absorb the triangular factors into the next core. Thus the $(r_{k-1}p_k)\times r_k$ matrix associated with G_k has orthonormal columns for $k < K$, and $\|G_K\|_F = \|A\|_F$. These steps use no inverse lower bound on a singular value.

Expand the contraction over all internal paths $(a_1, \dots, a_{K-1})$. A path contributes the product tensor with physical vectors $G_k[a_{k-1}, :, a_k]$. There are at most $\prod_{k<K} r_k \le J$ paths. To sum their squared weights, let

$$t_k(b) = \sum_{a_1,\dots,a_{k-1}} \prod_{\ell=1}^{k} \|G_\ell[a_{\ell-1}, :, a_\ell]\|_2^2, \qquad a_k = b.$$

Column normalization gives $t_1(b) = 1$. If $t_{k-1}(a) = 1$ for all a, then

$$t_k(b) = \sum_a t_{k-1}(a)\|G_k[a, :, b]\|_2^2 = 1$$

for $k < K$. Hence the sum of all squared path weights is

$$\sum_a t_{K-1}(a)\|G_K[a, :, 1]\|_2^2 = \|G_K\|_F^2 = \|A\|_F^2.$$

The individual path terms need not be orthogonal; the prefix identity is what proves the required equality. □

A.2 Proof of Theorem 1

Roadmap. Apply subadditivity, modewise Hölder and Bessel, then control the component-weight sum.

Proof. Let $q = 2/K \le 1$ and use Lemma 1. Subadditivity of the qth power of absolute value gives

$$\sum_i |\langle A, V_i\rangle|^q \le \sum_j \sum_i \prod_{k=1}^{K} |\langle x_{kj}, v_{ki}\rangle|^{2/K}.$$

Hölder's inequality with K equal exponents, followed by Bessel's inequality for each orthonormal family, yields

$$\begin{aligned}\sum_i \prod_k |\langle x_{kj}, v_{ki}\rangle|^{2/K} &\le \prod_k \left(\sum_i |\langle x_{kj}, v_{ki}\rangle|^2\right)^{1/K} \\ &\le \prod_k \|x_{kj}\|_2^{2/K} = (a_j^{\mathrm{cp}})^q.\end{aligned}$$

Finally, the power-mean inequality (or Hölder with exponents $2/q$ and $2/(2-q)$) gives

$$\sum_j (a_j^{\mathrm{cp}})^q \le J_A^{1-q/2} \left\{\sum_j (a_j^{\mathrm{cp}})^2\right\}^{q/2} \le J^{1-q/2} M^q.$$

This is (14). For $J_A = 0$ both sides of the first inequality are zero. □

A.3 Exact rank-one diagonal image

Lemma 2 (Exact diagonal reduction)**.** *Under* (4)*, the image of the rank-one tensor Frobenius ball of radius M under $A \mapsto (\langle A, V_i\rangle)_{i=1}^m$ is exactly*

$$\left\{ a \in \mathbb{R}^m : \sum_i |a_i|^{2/K} \le M^{2/K} \right\}.$$

Roadmap. Use the diagonal bound for one inclusion and construct signed product factors for the converse.

Proof. One inclusion follows from Theorem 1 with $J = 1$. Conversely, for any vector in the displayed ball, put

$$x_1 = \sum_i \operatorname{sign}(a_i)|a_i|^{1/K} v_{1i}, \qquad x_k = \sum_i |a_i|^{1/K} v_{ki}, \quad k \ge 2,$$

and set $A = x_1 \otimes \cdots \otimes x_K$. Orthonormality gives $\langle A, V_i\rangle = a_i$ and

$$\|A\|_F = \prod_k \|x_k\|_2 = \left(\sum_i |a_i|^{2/K} \right)^{K/2} \le M.$$

The tensor has every Tucker and TT rank at most one. The zero vector is handled by the zero tensor. □

A.4 Threshold error and structural radius

Lemma 3 (Scalar threshold envelope)**.** *If $|e| \le \lambda$ and $s = \operatorname{soft}_\lambda(x + e)$, then s lies between zero and x and*

$$|s - x| \le \min(|x|, 2\lambda).$$

For $q \in (0, 1]$, $t, u \ge 0$, one also has $\min(t, u) \le t^q u^{1-q}$, with the continuous interpretation at zero.

Roadmap. Check positive and negative scalar inputs, then interpolate the two deterministic error bounds.

Proof. For $x \ge 0$, $x+e \ge -\lambda$, so thresholding cannot produce a negative value. Also $x+e-\lambda \le x$, so $0 \le s \le x$. If $x > 2\lambda$, then $x+e > \lambda$ and $x-s = \lambda-e \le 2\lambda$. Otherwise $x-s \le x \le 2\lambda$. The negative case follows by oddness. If $t \le u$, then $t = t^q t^{1-q} \le t^q u^{1-q}$; if $u \le t$, then $u = u^q u^{1-q} \le t^q u^{1-q}$. □

Lemma 4 (Matrix functional bound)**.** *If $\operatorname{rank}(B) \le r$ and H has singular values $s_1(H) \ge s_2(H) \ge \cdots$, then*

$$|\langle H, B\rangle| \le \|B\|_F \left(\sum_{i \le r} s_i(H)^2 \right)^{1/2}.$$

Roadmap. Project onto the matrix column space and maximize the spectral trace over projections of bounded rank.

Proof. Let P project onto the column space of B. Then $PB = B$ and Cauchy–Schwarz gives $|\langle H, B\rangle| = |\langle PH, B\rangle| \le \|PH\|_F \|B\|_F$. Write $HH^\top = \sum_i \lambda_i u_i u_i^\top$, with decreasing nonnegative eigenvalues. The numbers $b_i = u_i^\top P u_i$ satisfy $0 \le b_i \le 1$ and $\sum_i b_i = \operatorname{tr}(P) \le r$. Therefore $\|PH\|_F^2 = \sum_i \lambda_i b_i \le \sum_{i \le r} \lambda_i$: moving any weight from a smaller eigenvalue to an unfilled larger one cannot reduce this sum. Since $\lambda_i = s_i(H)^2$, the result follows. □

A.5 Proof of Theorem 2

Roadmap. Construct a uniform coordinate-noise event, bound the threshold error pathwise, verify the other intervals and use the deterministic selection rule.

Proof. The product atoms are orthonormal because

$$\langle V_i, V_j \rangle = \prod_k \langle v_{ki}, v_{kj} \rangle = \mathbf{1}\{i = j\}.$$

Gaussian isotropy then gives the independent normals in (5). For a standard normal g, Chernoff's inequality gives $\mathbb{P}(g > t) \le \exp(-t^2/2)$ by minimizing $\exp(-ut + u^2/2)$ at $u = t$. Apply the same inequality to $-g$ and take a union bound to obtain

$$\mathbb{P}\{\max_i |\tau g_i| > \lambda\} \le 2m \exp\{-\lambda^2/(2\tau^2)\} = \alpha.$$

On the complementary event, Lemma 3 and Theorem 1 give

$$\begin{aligned} |\widehat{\psi}_{\mathrm{NL}} - \psi(A)| &\le \|w\|_\infty \sum_i \min(|a_i|, 2\lambda) \\ &\le \|w\|_\infty (2\lambda)^{1-q} \sum_i |a_i|^q \le h_{\mathrm{NL}}. \end{aligned}$$

This proves the uniform coverage of I_{NL}.

Because $\|L\|_F = 1$, $\langle L, Z \rangle - \psi(A) \sim N(0, \tau^2)$, so I_G has exact coverage. For any constrained unfolding, the loading matrix has singular values $|w_i|$: the product vectors on each side form orthonormal families. Lemma 4 thus gives $|\psi(A)| \le M(\sum_{i \le r} w_{(i)}^2)^{1/2}$. Choosing the smallest cap proves $|\psi(A)| \le h_0$, so I_0 has coverage one. The selected index in (13) is deterministic, hence the selected interval retains its individual coverage without a multiple-selection correction. Its length is exactly (15). □

A.6 Proof of Theorem 3

Roadmap. Force a variance cost at zero and a bias cost at signed coordinate tensors, then combine the two necessary inequalities.

Proof. Set $t = \langle W, L \rangle$. At $A = 0$ the center error is normal with mean b and standard deviation $v = \tau \|W\|_F$. For fixed h and $v > 0$, the probability that this variable belongs to $[-h, h]$ is maximized at $b = 0$. Indeed, for $b \ge 0$ its derivative is

$$v^{-1}\{\phi((h + b)/v) - \phi((h - b)/v)\} \le 0,$$

since $|h+b| \ge |h-b|$; the function is even in b. Honesty at zero therefore implies $h \ge z\tau \|W\|_F \ge z\tau |t|$. If $v = 0$, the same last inequality is automatic.

At $A = \pm M E_i$, the mean center error is

$$b \pm M(\langle W, E_i \rangle - p^{-1/2}).$$

At least one sign has absolute mean at least $M|\langle W, E_i \rangle - p^{-1/2}|$. A symmetric interval whose radius is strictly smaller than this mean magnitude has probability below $1/2$ under a nondegenerate

normal distribution: it lies strictly to one side of the median. In the degenerate case the coverage is zero. Since $1-\alpha > 1/2$, honesty forces

$$h \geq M \max_i |\langle W, E_i\rangle - p^{-1/2}| \geq \frac{M}{\sqrt{p}}|1-t|.$$

The last inequality follows from $t - 1 = p^{-1/2}\sum_i(\langle W, E_i\rangle - p^{-1/2})$. Put $a = M/\sqrt{p}$ and $b' = z\tau$. The two bounds imply

$$1 \leq |1-t| + |t| \leq h/a + h/b',$$

which rearranges to (17). On the rank-one tensor ball, $h_0 = M/\sqrt{p}$, and the direct/structural choice gives the matching order. No result about arbitrary data-dependent widths is invoked. □

A.7 A mixture inequality for confidence intervals

Lemma 5 (Common-target alternatives)**.** *Let P_0 have target zero and let Q be a mixture of alternatives whose targets all equal $\delta > 0$. If C covers zero under P_0 and covers the target at every alternative with probability at least $1-\alpha$, then*

$$\mathbb{E}_0|C| \geq \delta\{1 - 2\alpha - \mathrm{TV}(P_0, Q)\}. \tag{32}$$

If $Q \ll P_0$, then $\mathrm{TV}(P_0, Q) \leq \frac{1}{2}\sqrt{\chi^2(Q, P_0)}$.

Roadmap. Transfer coverage from the common-target mixture to the null using total variation, then intersect the two coverage events.

Proof. Mixture coverage gives $Q(\delta \in C) \geq 1-\alpha$. By the definition of total variation, $P_0(\delta \in C) \geq 1 - \alpha - \mathrm{TV}(P_0, Q)$. Intersecting with $\{0 \in C\}$ gives probability at least $1 - 2\alpha - \mathrm{TV}$. An interval containing both points has length at least δ, proving (32). The divergence bound follows from Cauchy–Schwarz applied to $\frac{1}{2}\mathbb{E}_0|dQ/dP_0 - 1|$. Appending an independent random seed leaves total variation unchanged, so randomized intervals are covered as well. □

For Gaussian shifts $P_B = N(\mathrm{vec}(B), \tau^2 I_D)$, direct integration of their likelihood ratios gives

$$\mathbb{E}_0\left[\frac{dP_B}{dP_0}\frac{dP_C}{dP_0}\right] = \exp\{\langle B, C\rangle/\tau^2\}. \tag{33}$$

To verify this, each ratio is $\exp\{\langle Z, B\rangle/\tau^2 - \|B\|_F^2/(2\tau^2)\}$. Multiply the ratios and use the moment generating function of $\langle Z, B+C\rangle$, whose variance under P_0 is $\tau^2\|B+C\|_F^2$. The exponent simplifies to (33).

A.8 Proof of Theorem 4

Roadmap. Construct rank-one matrix alternatives within the chosen unfolding and bound their full Gaussian likelihood second moment using an auxiliary Gaussian.

Proof. For the given bipartition, let $f_i = \bigotimes_{k\in S} e_i$ and $g_i = \bigotimes_{k\notin S} e_i$. Both families are orthonormal. Let $u = p^{-1/2}(s_1, \ldots, s_p)$ with independent uniform signs, and define

$$\theta = \min(M, \tau\sqrt{p}/4), \qquad A_u = \theta\left(\sum_i u_i f_i\right) \otimes \left(\sum_j u_j g_j\right).$$

Every alternative has rank one in this unfolding, norm θ, and functional value $\delta = \theta/\sqrt{p}$. Let Q average their Gaussian laws. If v is an independent copy of u, then (33) gives

$$1 + \chi^2(Q, P_0) = \mathbb{E}_{u,v} \exp\{\theta^2 (u^\top v)^2/\tau^2\} = \mathbb{E} \exp\{a S_p^2/p\},$$

where S_p is a sum of p independent signs and $a = \theta^2/(\tau^2 p) \le 1/16$.

Introduce an independent standard normal H. Its moment generating function yields

$$\begin{aligned}\mathbb{E} \exp(a S_p^2/p) &= \mathbb{E}_H \left\{ \cosh(\sqrt{2a/p}\, H) \right\}^p \\ &\le \mathbb{E}_H \exp(aH^2) = (1-2a)^{-1/2} \le \sqrt{8/7}.\end{aligned}$$

The inequality uses $\log\cosh(x) \le x^2/2$: for $x \ge 0$, its derivative $\tanh(x)$ is at most x, and both functions are even. The Gaussian-square expectation follows by integrating $\exp\{-(1-2a)h^2/2\}/\sqrt{2\pi}$; it is finite since $a < 1/2$. Nonnegative integrands justify exchanging the expectations.

Lemma 5 now gives (19). The constant is positive for $\alpha < 1/4$. On Θ_S, Lemma 4 with loading singular values all equal to $p^{-1/2}$ gives $|\psi(A)| \le M/\sqrt{p}$. Choosing between this structural interval and the direct interval attains the lower order. All calculations use the entire Gaussian experiment; the result does not arise from hiding off-diagonal observations. □

A.9 Proof of Theorem 5

Roadmap. Substitute the specified dimensions and norm scaling, then apply the two lower-bound theorems with their confidence-level restrictions.

Proof. Substitute $M = \tau\sqrt{p}$, $q = 2/K$ and $\|w\|_\infty = p^{-1/2}$ into (10). This gives

$$h_{\mathrm{NL}} = \tau J^{1-1/K} p^{1/K - 1/2} \{8 \log(2p/\alpha)\}^{(K-2)/(2K)},$$

which is the radius in (20). For fixed $K \ge 3$, its exponent $\beta = (K-2)/(2K)$ is positive and $(\log p/p)^\beta \to 0$. The cap classes contain all $\pm M E_i$ and zero, so Theorem 3 gives radius at least $z\tau/(1+z)$. For any retained constrained unfolding, the corresponding matrix-rank cap class contains Θ_S, and Theorem 4 gives expected length at zero at least $c_\alpha \tau/4$. The same comparisons under the stated growing-cap condition follow by retaining the factor $J^{1-1/K}$. □

A.10 Proof of Theorem 6

Roadmap. Mix orthogonal coordinate shifts, compute the full likelihood second moment exactly and apply the interval mixture lemma.

Proof. Let $a = \min\{M, \tau\sqrt{\log(1+p/4)}\}$ and mix uniformly over $A_j = aE_j$, $j = 1, \dots, p$. Each is in the rank-one tensor class and has target $\delta = a/\sqrt{p}$. Orthogonality of the E_j and (33) imply

$$1 + \chi^2(Q, P_0) = p^{-2}\{p \exp(a^2/\tau^2) + p(p-1)\}.$$

Thus $\chi^2(Q, P_0) = \{\exp(a^2/\tau^2) - 1\}/p \le 1/4$ and $\mathrm{TV}(P_0, Q) \le 1/4$. Lemma 5 gives (21). This construction does not establish that these alternatives are least favorable. □

A.11 Proof of Proposition 1

Roadmap. Construct sparse diagonals inside the rank-one tensor class, bound the diagonal likelihood mixture using factorial moments and balance the support size. The full-data likelihood is kept separate.

Proof. For an integer $1 \le s \le p$, let S be a uniformly chosen size-s subset of $\{1, \dots, p\}$. Define

$$A_S = b_s \left(\sum_{i \in S} e_i \right)^{\otimes K}.$$

This is a rank-one tensor, $\|A_S\|_F = b_s s^{K/2} \le M$, and every alternative has target $\delta_s = s b_s / \sqrt{p}$. Its observed diagonal has mean $b_s \mathbf{1}_S$. Let Q_d be the uniform mixture of these diagonal Gaussian laws, and let $P_{0,d}$ be the zero-mean diagonal law. Applying the Gaussian shift calculation (33) in dimension p gives

$$1 + \chi^2(Q_d, P_{0,d}) = \mathbb{E}_{S,T} \exp\{b_s^2 |S \cap T| / \tau^2\}, \tag{34}$$

where S, T are independent uniform size-s subsets. This is the sparse-mixture mechanism used in classical sequence lower bounds [Collier et al., 2017]; the following counting argument records the needed bound explicitly.

Conditional on S, $H = |S \cap T|$ is hypergeometric with population size p, success count s and sample size s. Write $(x)_j = x(x-1)\cdots(x-j+1)$, with $(x)_0 = 1$. Counting the j-element subsets of S contained in T yields

$$\mathbb{E}\binom{H}{j} = \binom{s}{j} \frac{(s)_j}{(p)_j} \le \binom{s}{j} (s/p)^j, \qquad 0 \le j \le s.$$

For $j \ge 1$, the inequality follows factor by factor from $(s-\ell)/(p-\ell) \le s/p$ for $0 \le \ell < j$, since $s \le p$. Set $t = \exp(b_s^2/\tau^2) - 1 \ge 0$. Expanding $(1+t)^H$ and using the preceding factorial moments gives

$$\begin{aligned} \mathbb{E} \exp(b_s^2 H / \tau^2) &= \sum_{j=0}^{s} t^j \mathbb{E}\binom{H}{j} \\ &\le (1 + st/p)^s \le \exp(s^2 t / p) \le e^{1/4}. \end{aligned}$$

The last inequality uses the definition of b_s. Equation (34) and Cauchy–Schwarz therefore give

$$\mathrm{TV}(Q_d, P_{0,d}) \le \frac{1}{2} \sqrt{e^{1/4} - 1}.$$

Lemma 5, now applied to the diagonal experiment, proves (23). The constant is positive because $\alpha < 1/4$ and $\frac{1}{2}\sqrt{e^{1/4}-1} < 1/2$. The argument also covers independent randomization.

To prove (24), let $x_p = (p/\log p)^{1/K}$ and $s_p = \lfloor x_p \rfloor$. For sufficiently large p, $x_p/2 \le s_p \le x_p$ and $s_p \ge 1$. When $M = \tau\sqrt{p}$, the norm term in b_{s_p} is at least $\tau\sqrt{\log p}$. The noise term is at most this quantity, since $1 + p/(4s_p^2) \le 1 + p/4 \le p$ for $p \ge 2$. Thus the minimum equals the noise term. Since $K > 2$ and $s_p \le p^{1/K}$ eventually,

$$\log\{1 + p/(4s_p^2)\} \ge \log\{1 + p^{1-2/K}/4\} \ge \tfrac{1}{2}(1 - 2/K) \log p$$

for sufficiently large p. Consequently,

$$\frac{s_p b_{s_p}}{\sqrt{p}} \geq \frac{1}{2}\sqrt{\frac{1-2/K}{2}}\,\tau\left(\frac{\log p}{p}\right)^{(K-2)/(2K)}.$$

Theorem 2 with $J = 1$ gives an upper bound of the same order and deterministic length. Since the worst-case criterion is at least the criterion at zero, both rates follow. The fixed-cap extension follows by class inclusion for the lower bound and a constant $J^{1-1/K}$ for the upper bound.

For completeness, the corresponding *full-tensor* mixture would have $\langle A_S, A_T\rangle = b_s^2|S\cap T|^K$ in (33). Its likelihood second moment is not (34), and the preceding bound does not apply to it. The proposition is restricted to diagonal observations throughout. □

B Proofs of the full-data refinements

Lemma 6 (Finite product nets and injective norm)**.** *The nets used in* (25) *exist and satisfy* $\rho(B) \leq 2N_S(B)$ *for any tensor on* S^K*. Furthermore, a tensor* B *in either rank-cap class satisfies* $\|B\|_F \leq \sqrt{J}\,\rho(B)$*.*

Roadmap. Use a volume bound and product telescoping, then apply the canonical squared-weight identity.

Proof. A maximal ε-separated subset of the unit sphere is an ε-net. Balls of radius $\varepsilon/2$ around its points are disjoint and contained in the ball of radius $1+\varepsilon/2$. The ratio of their Euclidean volumes bounds the cardinality by $(1+2/\varepsilon)^s$. Compactness gives a finite maximal set. Take $\varepsilon = 1/(2K)$.

Approximate each factor of any unit product direction by a net point. Telescoping the product one factor at a time, each contraction difference has magnitude at most $\varepsilon\rho(B)$, by multilinearity and the definition of ρ. The sum is at most $K\varepsilon\rho(B) = \rho(B)/2$. Taking the supremum gives $\rho(B) \leq N_S(B) + \rho(B)/2$.

For the second assertion use Lemma 1, writing its product terms as X_j. Then

$$\|B\|_F^2 = \sum_j \langle B, X_j\rangle \leq \rho(B)\sum_j \|X_j\|_F \leq \rho(B)\sqrt{J}\left(\sum_j \|X_j\|_F^2\right)^{1/2} = \rho(B)\sqrt{J}\,\|B\|_F.$$

The middle inequality is Cauchy–Schwarz, and the last identity is the squared-weight conclusion of Lemma 1. Divide when $B \neq 0$; the zero case is immediate. □

B.1 Proof of Theorem 7

Roadmap. A finite Gaussian union bound controls every net contraction. Test acceptance on that event bounds each ordered diagonal coordinate. Summation gives the interval and intersection gives the two expected-length guarantees.

Proof. There are at most $(ep/s)^s$ subsets of size s, since $\binom{p}{s} \leq p^s/s!$ and $\log(s!) \geq \int_1^s \log x\,dx \geq s\log s - s$. Each unit product contraction of τG has distribution $N(0,\tau^2)$. Therefore the Gaussian Chernoff bound gives

$$\begin{aligned}\mathbb{P}\{\exists S : N_S(\tau G) > \lambda^{\mathrm{scan}}_{|S|}\} &\leq \sum_{s=1}^{p} 2(ep/s)^s(1+4K)^{Ks}\exp\{-(\lambda^{\mathrm{scan}}_s)^2/(2\tau^2)\}\\ &\leq \gamma.\end{aligned}$$

Let $\mathcal{E}_\gamma$ be the complementary event. No independence between the scanned contractions is required.

On $\mathcal{E}_\gamma$, if the test accepts, the triangle inequality for the finite maximum implies $N_S(A) \le 2\lambda_s^{\mathrm{scan}}$. Lemma 6 gives $\rho(A_{S,\dots,S}) \le 4\lambda_s^{\mathrm{scan}}$. Coordinate restriction cannot increase any constrained unfolding rank, because it multiplies an unfolding by selection matrices. Theorem 1 and Lemma 6 thus imply

$$\sum_{i\in S} |A_{i,\dots,i}|^q \le J^{1-q/2}\|A_{S,\dots,S}\|_F^q \le J\,\rho(A_{S,\dots,S})^q.$$

Let $|a|_{(s)}$ be the sth largest absolute diagonal entry. Choose S to contain the s largest entries. It follows that

$$s|a|_{(s)}^q \le J(4\lambda_s^{\mathrm{scan}})^q, \qquad |a|_{(s)} \le 4J^{K/2}\lambda_s^{\mathrm{scan}} s^{-K/2}.$$

Summing these inequalities and using $|\psi(A)| \le p^{-1/2}\sum_i |a_i|$ proves the bound $|\psi(A)| \le R_\gamma$ in (26). On rejection the structural interval covers surely. Therefore C_{scan} covers on $\mathcal{E}_\gamma$ and is honest at level $1-\gamma$.

Both intersected intervals cover with probability at least $1-\alpha_0-\gamma$. On that event the intersection is nonempty and covers; replacing empty intersections by $\{0\}$ preserves this argument. Its length is always at most ℓ_*. At zero, the rejection probability is at most γ, and on acceptance the intersection has length at most $2R_\gamma$. This gives (27), without independence between the intervals.

For the specified γ, elementary logarithm bounds give $\lambda_s^{\mathrm{scan}} \le C_K\tau\sqrt{s\log(2p/\alpha)}$. For example, use $s\log(ep/s) \le s(1+\log p)$, $\log(2p/\gamma) \le 2\log(2p/\alpha)$ and $s \ge 1$. Equation (26) now proves (28); the extra term $\ell_*\gamma$ is smaller in order by Theorem 5. If $K \ge 4$, the sum is at most $1 + 2/(K-3)$ by integration of $x^{-(K-1)/2}$ over $[1,\infty)$. If $K = 3$, it is at most $1+\log p$. Combining the upper bound for $K \ge 4$ with Theorem 6 proves (29). □

Lemma 7 (Uniform cubic moment for a weighted sign sum)**.** *For independent uniform signs ϵ_i, let $S_p = \sum_{i=1}^p i^{-2/3}\epsilon_i$. Then, uniformly in $p \ge 1$,*

$$\mathbb{P}(|S_p| > x) \le 2e^{-x^3/192} \quad (x \ge 12), \qquad \mathbb{E}\exp\{|S_p|^3/40000\} \le 20/19 + 1/100 < 5/4.$$

Roadmap. Split the weighted sign sum into a bounded head and a sub-Gaussian tail, then integrate its cubic tail probability.

Proof. For $x \ge 12$, put $m = \lfloor (x/6)^3 \rfloor$. If $m \ge p$, then $|S_p| \le 3p^{1/3} \le x/2$. Otherwise, the first m terms have absolute sum at most $3m^{1/3} \le x/2$. The remaining independent sign sum has variance proxy

$$V_m = \sum_{i>m} i^{-4/3} \le 3m^{-1/3} \le 24/x.$$

Here $m \ge (x/6)^3/2$ and $18\cdot 2^{1/3} < 24$. The product inequality $\prod_i \cosh(ti^{-2/3}) \le \exp(t^2V_m/2)$ and Chernoff optimization give a two-sided tail bound $2\exp\{-(x/2)^2/(2V_m)\} \le 2e^{-x^3/192}$.

Put $u = 1/40000$ and $Y = |S_p|^3$. Tonelli's theorem applied to $e^{uY} = 1 + \int_0^Y ue^{ut}\,dt$ yields

$$\mathbb{E}e^{uY} \le e^{1728u} + \frac{2u}{1/192-u}e^{-(1/192-u)1728} \le \frac{20}{19} + \frac{1}{100}.$$

The last step uses $1728u < 1/20$, $e^x \le (1-x)^{-1}$ for $0 \le x < 1$, and $1/192 - u \ge 1/200$. The exponential factor in the second term is at most one. These estimates are uniform in p. □

B.2 Proof of Theorem 8

Roadmap. Couple three rank-one factor signs so that the diagonal is deterministic and harmonic. Control the full likelihood mixture by the cubic moment lemma and Hölder, without assuming independence of the three inner-product sums.

Proof. For independent uniform sign vectors $u, v \in \{-1, 1\}^p$, define

$$(x_u)_i = u_i i^{-1/3}, \qquad A_{u,v} = \frac{\tau}{200} x_u \otimes x_v \otimes x_{uv},$$

where $(uv)_i = u_i v_i$. Every alternative is rank one and

$$\|A_{u,v}\|_F = \frac{\tau}{200} \left(\sum_i i^{-2/3} \right)^{3/2} \le \frac{3^{3/2}}{200} \tau \sqrt{p} \le M.$$

The sum bound follows from $1 + \int_1^p x^{-2/3}\, dx \le 3p^{1/3}$. Its diagonal is $\tau/(200i)$, so the common target is $\delta = \tau H_p/(200\sqrt{p})$.

Let Q mix the full Gaussian laws over u, v. For independent copies u', v', set $\epsilon_i = u_i u'_i$ and $\eta_i = v_i v'_i$. The full tensor inner product divided by τ^2 is $S_1 S_2 S_3/40000$, where

$$S_1 = \sum_i i^{-2/3} \epsilon_i, \quad S_2 = \sum_i i^{-2/3} \eta_i, \quad S_3 = \sum_i i^{-2/3} \epsilon_i \eta_i.$$

Each has the marginal distribution in Lemma 7. They are dependent. The likelihood identity (33), the arithmetic–geometric mean inequality and Hölder give, with $u_0 = 1/40000$,

$$\begin{aligned} 1 + \chi^2(Q, P_0) &= \mathbb{E} e^{u_0 S_1 S_2 S_3} \\ &\le \mathbb{E} e^{u_0 (|S_1|^3 + |S_2|^3 + |S_3|^3)/3} \\ &\le \prod_{j=1}^{3} \{ \mathbb{E} e^{u_0 |S_j|^3} \}^{1/3} < 5/4. \end{aligned}$$

Thus $\mathrm{TV}(Q, P_0) < 1/4$. Lemma 5 gives $\mathbb{E}_0 |C| \ge (3/4 - 2\alpha)\delta$, proving (30). Since $H_p \asymp \log p$, this also proves the claimed obstruction to the earlier zero-point conjecture. □

C Supporting refinements

These separately calibrated extensions do not change the observation model of a lower bound by implication. The simulations retain the original Gaussian procedure.

C.1 Diagonal geometry

Proposition 2 (Smaller rank coefficient)**.** *In $\sum_i |a_i|^{2/K} \le D M^{2/K}$ one may take*

$$D_{\mathrm{Tu}} = \min \left\{ J_{\mathrm{Tu}}^{1-1/K}, m^{1-1/K}, (\prod_k \bar{r}_k)^{1/K} \right\}.$$

For TT caps put $\bar{r}_0 = \bar{r}_K = 1$ and $\rho_k = \min(p_k, \bar{r}_{k-1} \bar{r}_k)$; one may take

$$D_{\mathrm{TT}} = \min \left\{ J_{\mathrm{TT}}^{1-1/K}, m^{1-1/K}, (\prod_k \rho_k)^{1/K} \right\}.$$

The nonlinear radius remains valid with coefficient D.

Roadmap. Add ambient and mode-subspace bounds to the original expansion bound.

Proof. Bessel gives $\sum_i |a_i|^2 \le M^2$; Hölder in dimension m gives the second coefficient. For Tucker mode projectors P_k put $\ell_{ki} = \|P_k v_{ki}\|_2^2$. Projection leaves A unchanged, so Cauchy–Schwarz and Hölder give

$$\sum_i |a_i|^{2/K} \le M^{2/K} \sum_i \prod_k \ell_{ki}^{1/K} \le M^{2/K} \prod_k (\sum_i \ell_{ki})^{1/K} \le M^{2/K} \prod_k \bar{r}_k^{1/K}.$$

The last step uses $\sum_i v_{ki}^\top P_k v_{ki} \le \operatorname{tr} P_k$. For TT cores, fixing the two bond indices of core k expresses its physical fibers through at most $\bar{r}_{k-1}\bar{r}_k$ vectors. The mode rank is therefore at most ρ_k; the Tucker argument applies. Theorem 1 gives the first coefficient. □

For equal Tucker caps r this supplies coefficient r, instead of $r^{(K-1)^2/K}$; optimal rank dependence is not established. For unit loading vectors with frame bounds $\sum_i v_{ki} v_{ki}^\top \preceq F_k I$, the original proof instead gives coefficient $J^{1-1/K} \prod_k F_k^{1/K}$. This follows by replacing each Bessel inequality by its frame bound. The direct variance must use the actual $\|L\|_F^2$; the orthogonal structural formula is not asserted for these frames.

C.2 Exact affine benchmark

Let $Q_\alpha(\mu, v)$ be the smallest $h \ge 0$ such that $\mathbb{P}\{|N(\mu, v^2)| \le h\} \ge 1 - \alpha$, for $\mu, v \ge 0$, with $Q_\alpha(\mu, 0) = \mu$.

Proposition 3 (Structural range and fixed affine optimum)**.** *The range of ψ on either cap class under (4) is exactly $[-h_0, h_0]$. Under (16) and $0 < \alpha < 1/2$, set $a_0 = M\sqrt{s_{\mathrm{cap}}/p}$. Among all honest full-data fixed-radius affine intervals, the optimal radius is*

$$h^*_{\mathrm{aff}} = \min_{0 \le t \le 1} Q_\alpha(a_0(1-t), \tau t).$$

Center $t\langle L, Z\rangle$ attains the optimum. If $a_0 \le z_{1-\alpha}\tau$, then $h^*_{\mathrm{aff}} = a_0$.

Roadmap. Attain the structural bound, force bias and variance for any affine center, and reduce to a scalar optimization.

Proof. Let T index the s_{cap} largest weights. The tensor $M \sum_{i\in T} w_i V_i / (\sum_{i\in T} w_i^2)^{1/2}$ has norm M, all constrained ranks at most s_{cap}, and target h_0. Its signed scalar multiples attain the full interval; Lemma 4 gives the reverse inclusion.

For center $\langle W, Z\rangle + b$ put $t = \langle W, L\rangle$ and $g_i = \langle W, E_i\rangle - p^{-1/2}$. All s_{cap}-sparse diagonal tensors of norm at most M belong to the class. Opposite such tensors force an absolute mean error at least

$$|b| + M(\sum_{i=1}^{s_{\mathrm{cap}}} g_{(i)}^2)^{1/2} \ge a_0|1 - t|.$$

Indeed, the top-square sum is at least s_{cap}/p times the full sum, and $\|g\|_2 \ge |\sum_i g_i|/\sqrt{p} = |t - 1|$. The noise standard deviation is $\tau\|W\|_F \ge \tau|t|$. Confidence greater than one half forces h to be at least the absolute mean error, by the normal median argument. For $h \ge \mu \ge 0$ the probability $F(h; \mu, v) = \Phi((h-\mu)/v) + \Phi((h+\mu)/v) - 1$ decreases with μ and with $v > 0$. For μ, compare the normal densities. For v, its derivative is

$$-\{(h-\mu)\phi((h-\mu)/v) + (h+\mu)\phi((h+\mu)/v)\}/v^2 \le 0.$$

Thus $h \geq Q_\alpha(a_0|1-t|, \tau|t|)$; zero variance follows directly. For $t < 0$ this is at least a_0, and for $t > 1$ it is at least $z_{1-\alpha/2}\tau$. Both endpoints are achievable. On $[0,1]$, center $t\langle L, Z\rangle$ has exactly the asserted maximum bias and variance and attains the quantile bound. Continuity, including the zero-variance limit, gives a minimum. Finally $F(h;\mu,v) \leq \Phi((h-\mu)/v)$ implies $Q_\alpha(\mu,v) \geq \mu + z_{1-\alpha}v$. If $a_0 \leq z_{1-\alpha}\tau$, every $t \in [0,1]$ requires radius at least a_0, attained at $t = 0$. □

At cap one, $M = \tau\sqrt{p}$ and $\alpha = 0.05$, the exact affine length is 2τ. The scalar bounded-normal-mean mechanism is classical [Stark, 1992]; the tensor reduction is proved above.

C.3 Noise envelopes and approximate ranks

Define $\Theta^\eta = \{A : \exists B \in \Theta(M, \bar{r}),\ \|A - B\|_F \leq \eta\}$ for supplied $\eta \geq 0$.

Proposition 4 (Envelope and approximation calibration)**.** *Suppose $d_i = a_i + e_i$ and, uniformly over the asserted class, $\mathbb{P}(\max_i |e_i| \leq \lambda_0) \geq 1 - \alpha$. For any measurable $\Lambda \geq \lambda_0$, the threshold center has an honest radius*

$$\|w\|_\infty D M^{2/K} (2\Lambda)^{1-2/K} + \eta$$

on Θ^η. The structural radius is $h_0 + \eta$. Data-dependent selection between these two intervals is valid. A direct Gaussian alternative needs its own calibration and deterministic selection or joint error allocation.

Roadmap. Use one event for every allowed threshold and a residual Lipschitz bound.

Proof. On the envelope event, Lemma 3 holds simultaneously for all $\Lambda \geq \lambda_0$, proving the assertion at $\eta = 0$. For fixed e, the residual map $f_e(x) = \mathrm{soft}_\Lambda(x + e) - x = e - \mathrm{clip}(x + e, [-\Lambda, \Lambda])$ is one-Lipschitz. For $a_i = \langle A, V_i\rangle$ and $b_i = \langle B, V_i\rangle$,

$$|\sum_i w_i\{f_{e_i}(a_i) - f_{e_i}(b_i)\}| \leq \|w\|_2 \|a - b\|_2 \leq \|A - B\|_F.$$

The last step is Bessel. Add the threshold error at B. Also $|\psi(A) - \psi(B)| \leq \|L\|_F \|A - B\|_F \leq \eta$. Both intervals cover on the same event, proving the selection assertion. □

For iid Gaussian errors the exact threshold is $\tau\Phi^{-1}(\{1 + (1-\alpha)^{1/m}\}/2)$, by multiplying the m probabilities $2\Phi(\lambda_0/\tau) - 1$. Dependent Gaussian margins with variances at most τ^2 admit $\tau\Phi^{-1}(1 - \alpha/(2m))$ by Bonferroni. Marginal bounds $\mathbb{E}e^{te_i} \leq e^{t^2v^2/2}$ give $v\sqrt{2\log(2m/\alpha)}$ by Chernoff and a union bound. Bounds $\mathbb{E}|e_i|^a \leq v^a$ give $v(m/\alpha)^{1/a}$ by Markov and a union bound. Independence is unnecessary for these three union-bound calibrations.

Writing $\mathcal{D}_L A = (\langle A, V_i\rangle)_i$, the same proof gives

$$|\widehat{\psi} - \psi(A)| \leq \inf_B \left\{ \|w\|_\infty D_B \|B\|_F^{2/K} (2\Lambda)^{1-2/K} + \|\mathcal{D}_L(A - B)\|_2 \right\}.$$

The infimum ranges over tensors with their indicated valid budget D_B. Its unknown value is not an adaptive confidence radius.

For Tucker approximation let P_k be any leading mode singular-space projector of dimension $\min(\bar{r}_k, p_k)$ and put $B = (\bigotimes_k P_k)A$. The lifted projections commute. The identity $I - \prod_k P_k = \sum_k (\prod_{\ell<k} P_\ell)(I - P_k)$ has summands with mutually orthogonal ranges. Their squared residual norms add, each at most $\|(I - P_k)A\|_F^2$. Thus

$$\|A - B\|_F^2 \leq \sum_k \sum_{j > \bar{r}_k} \sigma_j(A_{(k)})^2, \qquad \|B\|_F \leq \|A\|_F.$$

This holds without a spectral gap, including ties. The distance-class proposition also covers TT approximation.

C.4 Finite variance and bounded row corruption

Proposition 5 (Replicate median calibration)**.** *Let clean tensor replicates be independent with common mean A, and every prescribed contraction have variance at most known v^2. Allow at most c adversarial replacements of entire replicates. Coordinates within a replicate may be dependent. Choose an odd block count b with*

$$b \geq 8c, \qquad b \geq 32\log(m/\alpha), \qquad n \geq 2b, \qquad k = \lfloor n/b \rfloor.$$

Partition the first bk replicates into fixed disjoint blocks. For coordinatewise medians d_i of the block means,

$$\mathbb{P}(\max_i |d_i - a_i| > 2v/\sqrt{k}) \leq \alpha.$$

Thus Proposition 4 applies. For the smallest admissible odd b, the threshold has order

$$O\{v\sqrt{c/n + \log(m/\alpha)/n}\}.$$

Roadmap. Bound clean-block errors, count corrupted blocks deterministically and concentrate the original bad-block count.

Proof. Chebyshev bounds a clean block's error probability at $2v/\sqrt{k}$ by $1/4$. For a fixed coordinate let I_j be its original clean-block error indicators. They are independent and $\sum_j \mathbb{E}I_j \leq b/4$. At most c summaries change even for an adaptive adversary. If the final median errs, $\sum_j I_j \geq b/2 - c \geq 3b/8$. For a Bernoulli I, $h(t) = \log \mathbb{E}e^{t(I-\mathbb{E}I)}$ satisfies $h(0) = h'(0) = 0$ and $h''(t) \leq 1/4$, the variance of a tilted Bernoulli. Hence $h(t) \leq t^2/8$. Independence and Markov give

$$\mathbb{P}\{\sum_j (I_j - \mathbb{E}I_j) \geq b/8\} \leq e^{-tb/8 + bt^2/8} = e^{-b/32} \quad (t = 1/2).$$

A union bound over m coordinates proves the claim. Since $k \geq n/(2b)$, the rate follows, including the fixed rounding constants in b. □

Median-of-means calibration is classical [Devroye et al., 2016]. For independent Bernoulli replacements of probability ϵ, let $c = \lceil 2n\epsilon + 2\log(1/\delta) \rceil$. Their count satisfies $\mathbb{E}2^N = (1+\epsilon)^n \leq e^{n\epsilon}$, giving $\mathbb{P}(N > c) \leq e^{n\epsilon - c\log 2} \leq \delta$. With envelope error budget α_0, coverage is at least $1 - \alpha_0 - \delta$ when the block conditions hold. These sufficient corruption conditions are not a sharp breakdown or contamination-minimax result.

C.5 Unknown Gaussian variance

Proposition 6 (Independent residual calibration)**.** *Retain $n \geq 2$ iid Gaussian full-tensor replicates with unknown noise standard deviation $\sigma > 0$. Let $d_{ji} = \langle W_j, V_i \rangle$, $\bar{d}_i = n^{-1}\sum_j d_{ji}$, $\nu = m(n-1)$, and*

$$S^2 = \nu^{-1} \sum_i \sum_j (d_{ji} - \bar{d}_i)^2, \qquad \sigma_U = S\sqrt{\nu/\chi^2_{\nu,\delta}}, \quad 0 < \delta < \alpha.$$

The denominator is the lower δ chi-square quantile. Compute the original three intervals with $\tau_U = \sigma_U/\sqrt{n}$ and $\alpha_0 = \alpha - \delta$, choosing the smallest using only S and supplied quantities. The selected interval is honest at level $1 - \alpha$. For fixed δ and $\nu \to \infty$, $\sigma_U/\sigma \to 1$ in probability, and the nonlinear expected-length order is preserved.

Roadmap. Orthogonally separate means and residuals, then condition on a conservative scale estimate.

Proof. An orthogonal transformation in the replicate index separates each sample mean from its $n-1$ residual coordinates. Joint Gaussianity and loading orthonormality give independence of all residual and mean coordinates. Therefore $\nu S^2/\sigma^2 \sim \chi^2_\nu$, independent of the vector of means, and $\mathbb{P}(\sigma_U \geq \sigma) = 1-\delta$. Conditional on S with $\sigma_U \geq \sigma$, both stochastic intervals have failure at most α_0 and the selected index is fixed. The structural interval covers surely. Integrating conditional failure and adding δ proves honesty.

The normalized chi-square has mean one and variance $2/\nu$. Chebyshev gives convergence to one and convergence of its fixed-δ quantile to one. Thus $\sigma_U/\sigma \to 1$ and $\mathbb{E}(\sigma_U/\sigma)^2 = \nu/\chi^2_{\nu,\delta} \to 1$. The second-moment bound gives uniform integrability for powers at most one. Since the nonlinear radius is proportional to $\tau_U^{1-2/K}$, its expected order is preserved. □

Retained residuals are essential to this construction; it is not an estimate of variance from Z alone.

C.6 Confidence levels and covariance

Proposition 7 (Calibrated lower bounds)**.** *The fixed-confidence rates of Theorems 6, 4 and Proposition 1 extend to every $0 < \alpha < 1/2$ after changing constants. If known Gaussian covariance satisfies $\Sigma \succeq s_0^2 I$ with $s_0 > 0$, all-procedure isotropic lower bounds transfer with $\tau = s_0$.*

Roadmap. Tune the mixture distance below $1-2\alpha$; compare covariance models by adding independent noise.

Proof. Fix $0 < \eta < 1-2\alpha$. For the coordinate mixture use $a = \min\{M, \tau\sqrt{\log(1+4\eta^2 p)}\}$, giving $\chi^2 \leq 4\eta^2$. For the unfolding mixture use $\theta = \min(M, c_\eta \tau\sqrt{p})$ with $c_\eta^2 = \{1-(1+4\eta^2)^{-2}\}/2$. Its second-moment bound is at most $1+4\eta^2$. For the diagonal support mixture use

$$b_s = \min\{Ms^{-K/2},\ \tau\sqrt{\log[1+\log(1+4\eta^2)p/s^2]}\}.$$

The factorial-moment bound again gives second moment at most $1+4\eta^2$. Total variation is at most η in each case; the interval lemma has positive factor $1-2\alpha-\eta$. The fixed-α rates are unchanged.

Add independent $N(0, \Sigma - s_0^2 I)$ noise to $N(A, s_0^2 I)$. Applying any Σ-procedure to the result gives an isotropic randomized procedure with the same coverage and length distributions. The isotropic all-procedure lower bound therefore applies. □

Upper envelope calibration uses the marginal variances $\mathrm{vec}(V_i)^\top \Sigma\, \mathrm{vec}(V_i)$; the direct radius uses $z_{1-\alpha/2}\sqrt{\mathrm{vec}(L)^\top \Sigma\, \mathrm{vec}(L)}$. A lower covariance bound is needed for a universal positive noise lower scale: if $\Sigma = \tau^2(I - \mathrm{vec}(L)\,\mathrm{vec}(L)^\top)$, the target is observed without noise. Cross terms $B^\top \Sigma^{-1} C$ cannot in general be ordered by a covariance bound; the randomization proof avoids that invalid step.

D Result status and remaining questions

Here **PROVED** denotes a complete supplied derivation, with finite numerical checks where useful. It does not denote independent peer review or machine-checked verification. Classical results used as methodological background are credited explicitly.

At $M = \tau\sqrt{p}$, fixed positive caps and fixed confidence, the unresolved cubic zero-point rate lies between $\tau \log p/\sqrt{p}$ and $\tau(\log p)^{3/2}/\sqrt{p}$. For the worst-case criterion, the present upper bound remains the diagonal order $\tau(\log p/p)^{(K-2)/(2K)}$, and the zero-point lower bounds need not be sharp

Item	Scope	Status
Theorems 1–6	Original geometry, coverage and comparison bounds; confidence-level domain corrected in Theorem 5	PROVED
Proposition 1	Classical diagonal-information minimax rate	PROVED
Theorem 7	Full-data scan; optimal expected length at zero for fixed $K \geq 4$	PROVED
Theorem 8	Cubic harmonic lower bound	PROVED
Appendix C	Improved geometry, affine optimum, calibrated noise and approximation extensions	PROVED
Q1a	Full-data expected length at zero for $K = 3$: logarithmic gap	PARTIAL
Q1b	Full-data worst-case expected length	PARTIAL
Original Q2	Coordinate lower-scale attainment for all $K \geq 3$	REFUTED
Q3	Actual-rank adaptation and arbitrary loading geometry	OPEN
Q4	Efficient computation attaining the scan guarantee	OPEN

Table 6: Revised ledger. Original Q2 is false for $K = 3$ and its rate is attained for fixed $K \geq 4$. None of the remaining open statements is used by a proved result.

for that larger criterion. The noise extensions require the observations and calibration explicitly stated in their propositions. No real-data or LLM validation is asserted. The priority of the new full-data constructions requires further independent literature and mathematical review.

References

Joshua Agterberg and Anru R. Zhang. Statistical Inference for Low-Rank Tensors: Heteroskedasticity, Subgaussianity, and Applications, 2024. URL https://arxiv.org/abs/2410.06381.

Timothy B. Armstrong, Martin Weidner, and Andrei Zeleneev. Robust Estimation and Inference in Panels with Interactive Fixed Effects. *Journal of Political Economy*, 2026. doi: 10.1086/741623. URL https://arxiv.org/abs/2210.06639. In press; accepted-paper record verified. Full text examined: arXiv v3, December 2024.

Anna Bykhovskaya, Vadim Gorin, and Sasha Sodin. How weak are weak factors? Uniform inference for signal strength in signal plus noise models, 2025. URL https://arxiv.org/abs/2507.18554.

Changxiao Cai, H. Vincent Poor, and Yuxin Chen. Uncertainty Quantification for Nonconvex Tensor Completion: Confidence Intervals, Heteroscedasticity and Optimality. *IEEE Transactions on Information Theory*, 69(1):407–452, 2023. doi: 10.1109/TIT.2022.3205781. URL https://arxiv.org/abs/2006.08580.

T. Tony Cai and Mark G. Low. An adaptation theory for nonparametric confidence intervals. *The Annals of Statistics*, 32(5):1805–1840, 2004a. doi: 10.1214/009053604000000049. URL https://arxiv.org/abs/math/0503662.

T. Tony Cai and Mark G. Low. Minimax estimation of linear functionals over nonconvex parameter spaces. *The Annals of Statistics*, 32(2):552–576, 2004b. doi: 10.1214/009053604000000094. URL https://arxiv.org/abs/math/0406427.

T. Tony Cai, Tengyuan Liang, and Alexander Rakhlin. Geometric inference for general high-dimensional linear inverse problems. *The Annals of Statistics*, 44(4):1536–1563, 2016. doi: 10.1214/15-AOS1426. URL https://arxiv.org/abs/1404.4408.

Alexandra Carpentier and Arlene K. H. Kim. An iterative hard thresholding estimator for low rank matrix recovery with explicit limiting distribution. *Statistica Sinica*, 28:1371–1393, 2018. doi: 10.5705/ss.202016.0103. URL https://arxiv.org/abs/1502.04654.

Alexandra Carpentier and Richard Nickl. On signal detection and confidence sets for low rank inference problems. *Electronic Journal of Statistics*, 9(2):2675–2688, 2015. doi: 10.1214/15-EJS1087. URL https://arxiv.org/abs/1507.03829.

Alexandra Carpentier, Olga Klopp, Matthias Löffler, and Richard Nickl. Adaptive confidence sets for matrix completion. *Bernoulli*, 24(4A):2429–2460, 2018. doi: 10.3150/17-BEJ933. URL https://arxiv.org/abs/1608.04861.

Jungjun Choi, Hyukjun Kwon, and Yuan Liao. Inference for Low-Rank Models Without Estimating the Rank. *Journal of the American Statistical Association*, 121(553):655–666, 2026. doi: 10.1080/01621459.2025.2538272. URL https://arxiv.org/abs/2311.16440.

Olivier Collier, Laëtitia Comminges, and Alexandre B. Tsybakov. Minimax estimation of linear and quadratic functionals on sparsity classes. *The Annals of Statistics*, 45(3):923–958, 2017. URL https://arxiv.org/abs/1502.00665.

Lieven De Lathauwer, Bart De Moor, and Joos Vandewalle. A multilinear singular value decomposition. *SIAM Journal on Matrix Analysis and Applications*, 21(4):1253–1278, 2000. doi: 10.1137/S0895479896305696. URL https://doi.org/10.1137/S0895479896305696.

Luc Devroye, Matthieu Lerasle, Gábor Lugosi, and Roberto I. Oliveira. Sub-gaussian mean estimators. *The Annals of Statistics*, 44(6):2695–2725, 2016. doi: 10.1214/16-AOS1440.

David L. Donoho. Statistical Estimation and Optimal Recovery. *The Annals of Statistics*, 22(1):238–270, 1994. doi: 10.1214/aos/1176325367. URL https://web.stanford.edu/dept/statistics/cgi-bin/donoho/wp-content/uploads/2018/08/SEOR.pdf.

David L. Donoho and Iain M. Johnstone. Ideal spatial adaptation by wavelet shrinkage. *Biometrika*, 81(3):425–455, 1994. doi: 10.1093/biomet/81.3.425. URL https://doi.org/10.1093/biomet/81.3.425.

Mathias Drton. Likelihood ratio tests and singularities. *The Annals of Statistics*, 37(2):979–1012, 2009. doi: 10.1214/07-AOS571. URL https://arxiv.org/abs/math/0703360.

Jicong Fan, Lijun Ding, Chengrun Yang, Zhao Zhang, and Madeleine Udell. Euclidean-norm-induced schatten-p quasi-norm regularization for low-rank tensor completion and tensor robust principal component analysis. *Transactions on Machine Learning Research*, 2023. URL https://arxiv.org/abs/2012.03436.

Shmuel Friedland and Lek-Heng Lim. Nuclear norm of higher-order tensors. *Mathematics of Computation*, 87(311):1255–1281, 2018. doi: 10.1090/mcom/3239. URL https://arxiv.org/abs/1410.6072.

Bin Gao, Renfeng Peng, and Ya-xiang Yuan. Low-rank optimization on Tucker tensor varieties. *Mathematical Programming*, 214(1):357–407, 2025. doi: 10.1007/s10107-024-02186-w. URL https://arxiv.org/abs/2311.18324.

Anatoli Juditsky and Arkadi Nemirovski. Near-optimal recovery of linear and N-convex functions on unions of convex sets. *Information and Inference: A Journal of the IMA*, 9(2):423–453, 2020. doi: 10.1093/imaiai/iaz011. URL https://arxiv.org/abs/1804.00355.

Baofang Ke, Zihao Song, Weihua Zhao, and Lei Wang. Efficient Inference for Low-rank Regression via Covariance-Weighted Projection. *Statistica Sinica*, 2026. doi: 10.5705/ss.202026.0222. URL https://www3.stat.sinica.edu.tw/statistica/fp/SS-2026-0222.html. Newly accepted preprint SS-2026-0222; updated September 8, 2026. No volume or pages assigned in inspected record.

Benjamin Kutschan. Tangent cones to tensor train varieties. *Linear Algebra and its Applications*, 544:370–390, 2018. doi: 10.1016/j.laa.2018.01.012. URL https://arxiv.org/abs/1705.10152.

Mark G. Low. On nonparametric confidence intervals. *The Annals of Statistics*, 25(6):2547–2554, 1997. doi: 10.1214/aos/1030741084. URL https://doi.org/10.1214/aos/1030741084.

Wanteng Ma, Lilun Du, Dong Xia, and Ming Yuan. Multiple testing of linear forms for noisy matrix completion, 2026. URL https://arxiv.org/abs/2312.00305. Version 3, June 30, 2026; tensor-completion lower bound in Theorem 10.

Ivan V. Oseledets. Tensor-train decomposition. *SIAM Journal on Scientific Computing*, 33(5): 2295–2317, 2011. doi: 10.1137/090752286. URL https://doi.org/10.1137/090752286.

Philip B. Stark. Affine minimax confidence intervals for a bounded normal mean. *Statistics & Probability Letters*, 13(1):39–44, 1992. doi: 10.1016/0167-7152(92)90234-V.

Larry Wasserman, Aaditya Ramdas, and Sivaraman Balakrishnan. Universal inference. *Proceedings of the National Academy of Sciences*, 117(29):16880–16890, 2020. doi: 10.1073/pnas.1922664117. URL https://arxiv.org/abs/1912.11436.

Dong Xia and Ming Yuan. Statistical inferences of linear forms for noisy matrix completion. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(1):58–77, 2021. doi: 10.1111/rssb.12400. URL https://arxiv.org/abs/1909.00116.

Dong Xia, Anru Zhang, and Yuchen Zhou. Inference for Low-rank Tensors—No Need to Debias. *The Annals of Statistics*, 50(2):1220–1245, 2022. doi: 10.1214/21-AOS2146. URL https://arxiv.org/abs/2012.14844.

Ke Xu, Elynn Chen, and Yuefeng Han. Statistical Inference for Low-Rank Tensor Models, 2025. URL https://arxiv.org/abs/2501.16223.

Yinchu Zhu. How well can we learn large factor models without assuming strong factors?, 2019. URL https://arxiv.org/abs/1910.10382.